

Advanced Web Technologies(AWT) Lab (MCAL25)**INDEX****Name of the faculty: Ganesh Bhagwat**

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7	Design a webpage to demonstrate the working of a simple stored procedure.		CO2	
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**PRACTICAL NO. 1: Design a Web Application for an Organization with
Registration forms and advanced controls.**

Webform.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="pract1.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server"> <title></title></head>
<body style="font-weight: 700">
  <form id="form1" runat="server">
    <div>
      Registration form<br />
      <br />
      First Name:
      <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
      <br />      <br />
      Last Name:
      <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
      <br />      <br />
      Email :
      <asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
      <br />      <br />
      Date of Birth :
      <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
      <br />      <br />
      Gender :
      <asp:DropDownList ID="DropDownList1" runat="server">
        <asp:ListItem>Select gender</asp:ListItem>
        <asp:ListItem>Male</asp:ListItem>
        <asp:ListItem>Female</asp:ListItem>
        <asp:ListItem>Other</asp:ListItem>
      </asp:DropDownList>
      <br />      <br />
      Department :
      <asp:RadioButtonList ID="RadioButtonList2" runat="server">
        <asp:ListItem>MCA</asp:ListItem>
        <asp:ListItem>MMS</asp:ListItem>
      </asp:RadioButtonList>
      <br />
      <asp:CheckBox ID="CheckBox1" runat="server" Text="I accept the terms and conditions" />
      <br />      <br />
      <asp:Button ID="Button1" runat="server" OnClick="Button1_Click" Text="Submit" />
    </div>
  </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace awtassignment
{
    public partial class WebForm3 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            if (CheckBox1.Checked)
            {
                string firstName = TextBox1.Text;
                string lastName = TextBox2.Text;
                string email = TextBox3.Text;
                string dob = TextBox4.Text;
                string gender = DropDownList1.SelectedValue;
                string department = RadioButtonList2.SelectedValue;

                // Display confirmation message
                Response.Write($"<h3>Registration Successful!</h3>");
                Response.Write($"<p>Name: {firstName} {lastName}</p>");
                Response.Write($"<p>Email: {email}</p>");
                Response.Write($"<p>Date of Birth: {dob}</p>");
                Response.Write($"<p>Gender: {gender}</p>");
                Response.Write($"<p>Department: {department}</p>");
            }
            else
            {
                Response.Write("<h3 style='color:red'>Please accept the terms and conditions.</h3>");
            }
        }
    }
}
```

Output:

Registration form

First Name:

Aniket

Last Name:

Shinde

Email :

123@123.com

Date of Birth :

18-06-2001

Gender :

Male

Department :

☒ MCA

☐ MMS

☒ I accept the terms and conditions

Submit

Registration Successful!

Name: Aniket Shinde

Email: 123@123.com

Date of Birth: 18-06-2001

Gender: Male

Department: MCA

PRACTICAL NO. 2: Create a website using the master page concept.**Webform.aspx:**

```
<%@ Page Title="" Language="C#" MasterPageFile="~/Site1.Master"
AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="AWT_prac.WebForm1" %>
<asp:Content ID="Content1" ContentPlaceHolderID="MainContent" runat="server">
    <h2>Registration Form</h2>
    <form id="form1" >
        <div>
            <label for="fullName">Full Name:</label>
            <asp:TextBox ID="txtFullName" runat="server" /><asp:RequiredFieldValidator
ID="rfvFullName" runat="server" ControlToValidate="txtFullName" ErrorMessage="Full
Name is required" ForeColor="Red" />
        </div>
        <div>
            <label for="email">Email:</label>
            <asp:TextBox ID="txtEmail" runat="server" />
            <asp:RegularExpressionValidator ID="revEmail" runat="server"
ControlToValidate="txtEmail"
            ValidationExpression="^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"
ErrorMessage="Enter a valid email address" ForeColor="Red" />
        </div>
        <div>
            <label for="password">Password:</label>
            <asp:TextBox ID="txtPassword" runat="server" TextMode="Password" />
            <asp:RegularExpressionValidator ID="revPassword" runat="server"
ControlToValidate="txtPassword"
            ValidationExpression="^(?=.*[A-Z])(?=.*\d){8,}$" ErrorMessage="Password
must be at least 8 characters, with one uppercase letter and one number" ForeColor="Red" />
        </div>
        <div>
            <label for="confirmPassword">Confirm Password:</label>
            <asp:TextBox ID="txtConfirmPassword" runat="server" TextMode="Password" />
            <asp:CompareValidator ID="cvConfirmPassword" runat="server"
ControlToValidate="txtConfirmPassword"
            ControlToCompare="txtPassword" ErrorMessage="Passwords do not match"
ForeColor="Red" />
        </div>
        <div>
            <label for="age">Age:</label>
            <asp:TextBox ID="txtAge" runat="server" />
            <asp:RangeValidator ID="rvAge" runat="server" ControlToValidate="txtAge"
MinimumValue="18" MaximumValue="60"
            Type="Integer" ErrorMessage="Age must be between 18 and 60" ForeColor="Red" />
        </div>
    </div>
</form>
</div>
```

```
<asp:CustomValidator ID="cvPasswordNameCheck" runat="server"
ControlToValidate="txtPassword"
    OnServerValidate="ValidatePassword" ErrorMessage="Password cannot contain
the user's name" ForeColor="Red" />
</div>
<div>
    <asp:ValidationSummary ID="ValidationSummary1" runat="server"
ForeColor="Red" />
</div>
<div>
    <asp:Button ID="btnRegister" runat="server" Text="Register"
OnClick="btnRegister_Click" />
</div>
</form>
</asp:Content>
```

Site1.master:

```
<%@ Master Language="C#" AutoEventWireup="true" CodeBehind="Site1.master.cs"
Inherits="AWT_prac.Site1" %>
<!DOCTYPE html>
<html>
<head runat="server">
    <title>ASP.NET Registration Form</title>
    <script src="https://code.jquery.com/jquery-3.6.0.min.js" type="text/javascript"></script>
    <script type="text/javascript">
        $(function () {$.validator.unobtrusive.parse('form');    });
    </script>
</head>
<body>
    <form id="form1" runat="server">
    <div class="header">
        <h1>Welcome to the Registration Page</h1>
    </div>
    <div class="navbar">
        <ul>
            <li><a href="Home.aspx">Home</a></li>
            <li><a href="About.aspx">About</a></li>
            <li><a href="Contact.aspx">Contact</a></li>
        </ul>
    </div>
    <div class="content">
        <asp:ContentPlaceHolder ID="MainContent"
runat="server"></asp:ContentPlaceHolder>
    </div>
    <div class="footer">
        <p>&copy; 2025 MCA-C24108. All Rights Reserved.</p>
    </div>
    </form>
</body>
</html>
```

Output:

Welcome to the Registration Page

[Home](#) [About](#) [Contact](#)

Registration Form

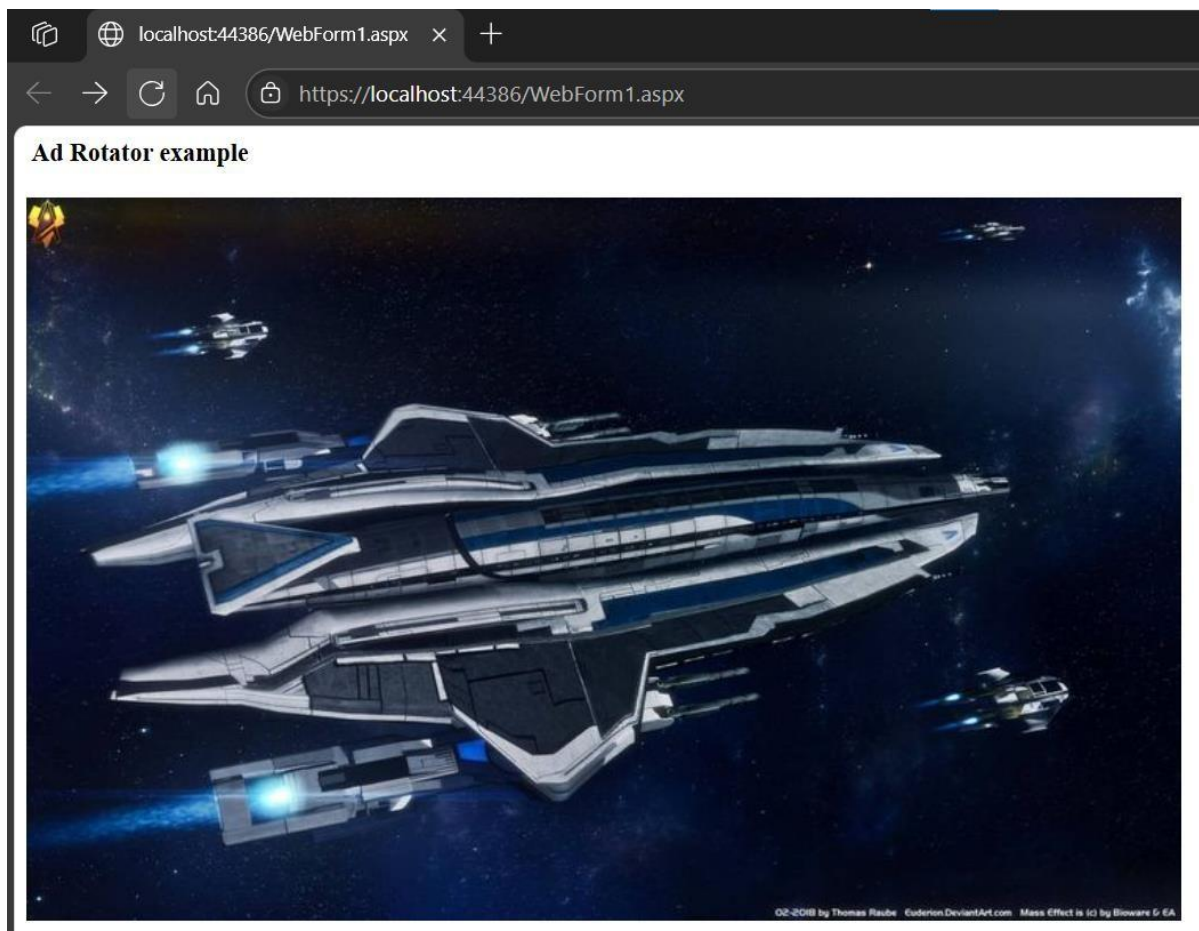
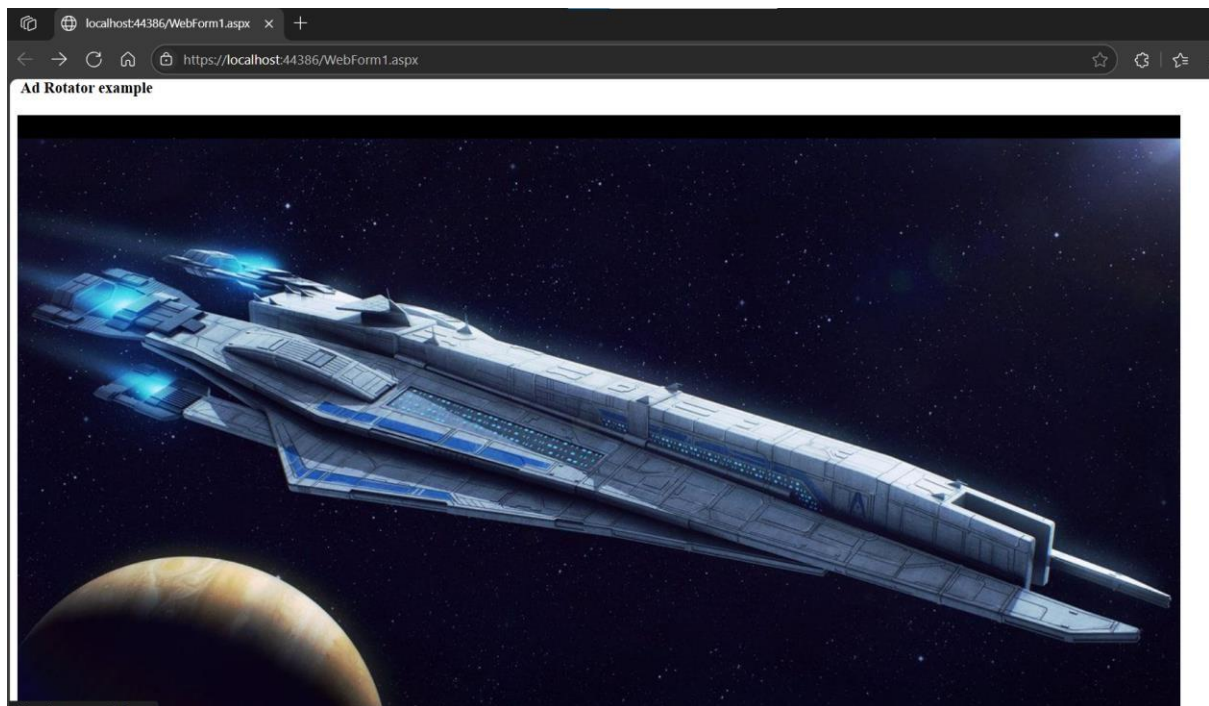
Full Name:	Aniket
Email:	123@123.com
Password:	...
	Password must be at least 8 characters, with one uppercase letter and one number
Confirm Password:	...
Age:	asa
	Age must be between 18 and 60
Register	

PRACTICAL NO. 3: Design a Web Application using advanced controls.**1. AdRotator Control in ASP.NET****Webform.aspx:**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication1.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body style="font-weight: 700">
    <form id="form1" runat="server">
        <div>
            &nbsp;Ad Rotator example<br />
            <br />
            <asp:AdRotator ID="AdRotator1" runat="server" DataSourceID="XmlDataSource"/>
            <asp:XmlDataSource ID="XmlDataSource" runat="server"
DataFile="~/XMLFile1.xml"></asp:XmlDataSource>
        </div>
    </form>
</body>
</html>
```

XMLFile1.xml:

```
<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
    <Ad>
        <ImageUrl>ssv bg.jpg</ImageUrl>
        <NavigateUrl>https://www.example2.com</NavigateUrl>
        <AlternateText>First Ad</AlternateText>
        <Impressions>50</Impressions>
    </Ad>
    <Ad>
        <ImageUrl>miranda-ssv bg.jpg</ImageUrl>
        <NavigateUrl>https://www.example1.com</NavigateUrl>
        <AlternateText>Second Ad</AlternateText>
        <Impressions>30</Impressions>
    </Ad>
</Advertisements>
```

Output:

2. Navigation Control (Menu Navigation)

Webform.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="WebApplication1.WebForm2" %>
```

```
<!DOCTYPE html>
<html>
<head runat="server">
<title>Navigation Example</title>
</head>
<body>
  <form id="form1" runat="server">
    <h2>ASP.NET Navigation Example</h2>
    <asp:Menu ID="Menu1" runat="server" Orientation="Horizontal">
      <Items>
        <asp:MenuItem Text="Home" NavigateUrl="Home.aspx"/>
        <asp:MenuItem Text="About" NavigateUrl="About.aspx"/>
        <asp:MenuItem Text="Contact" NavigateUrl="Contact.aspx"/>
      </Items>
    </asp:Menu>

  </form>
</body>
</html>
```

Output:



3. File Upload in ASP.NET

Webform.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="WebApplication1.WebForm2" %>
<!DOCTYPE html>
<html>
<head runat="server">
    <title>File Upload Example</title>
</head>
<body>
    <form id="form1" runat="server">
        <h2>Upload a File</h2>
        <asp:FileUpload ID="FileUploadControl" runat="server" />
        <asp:Button ID="UploadButton" runat="server" Text="Upload"
OnClick="UploadButton_Click" />
        <br />
        <asp:Label ID="StatusLabel" runat="server" Text=""></asp:Label>
    </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.IO;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace WebApplication1
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void UploadButton_Click(object sender, EventArgs e)
        {
            if (FileUploadControl.HasFile)
            {
                try
                {
                    string filename = Path.GetFileName(FileUploadControl.FileName);
                    FileUploadControl.SaveAs(Server.MapPath("~/Uploads/") + filename);
                }
            }
        }
    }
}
```

```
        StatusLabel.Text = "Upload status: File uploaded successfully!";
    }
    catch (Exception ex)
    {
        StatusLabel.Text = "Upload status: Error - " + ex.Message;
    }
}
else
{
    StatusLabel.Text = "Upload status: No file selected.";
}
}
}
```

Output:

Upload a File

C24108_AIML_Journal.pdf

**PRACTICAL NO. 4: Webpage Demonstrating Connection-Oriented Architecture
(ASP.NET Web Forms with SQL Server Database).****Webform.aspx:**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication2.WebForm1" %>

<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:Label ID="Label1" runat="server" ForeColor="#0033CC" Text="Aniket
Shinde- C24108"></asp:Label>
            <br />
            <br />
            <asp:Label ID="Label2" runat="server" Text="Connection Oriented
Architecture"></asp:Label>
            <br />
            <br />
            <asp:Button ID="Button1" runat="server" OnClick="Button1_Click" Text="Fetch
Data" />
            <br />
            <br />
            <asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="False"
DataSourceID="SqlDataSource1">
                <Columns>
                    <asp:BoundField DataField="StudentID" HeaderText="StudentID"
SortExpression="StudentID" />
                    <asp:BoundField DataField="StudentName" HeaderText="StudentName"
SortExpression="StudentName" />
                    <asp:BoundField DataField="Address" HeaderText="Address"
SortExpression="Address" />
                </Columns>
            </asp:GridView>
            <asp:SqlDataSource ID="SqlDataSource1" runat="server" ConnectionString="<%=
ConnectionStrings:StudentConnectionString %>" ProviderName="<%=
ConnectionStrings:StudentConnectionString.ProviderName %>" SelectCommand="SELECT
* FROM [Table_1]"></asp:SqlDataSource>
        </div>
    </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Net.Mail;
using System.Data.SqlClient;
using System.Xml.Linq;
using System.Data;

namespace WebApplication2
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            string connection = "Data Source=DESKTOP-TPEBNNC;Initial
Catalog=Student;Integrated Security=True";
            using (SqlConnection con = new SqlConnection(connection))
            {
                try
                {
                    con.Open();
                    string query = "SELECT * FROM users";
                    SqlDataAdapter da = new SqlDataAdapter(query, con);
                    DataTable dt = new DataTable();
                    da.Fill(dt);
                    GridView1.DataSource = dt;
                    GridView1.DataBind();
                }
                catch (Exception ex)
                {
                    Response.Write("<script>alert('Error: + "+ex.Message+"')</script>");
                }
            }
        }
    }
}
```

Output:

Aniket Shinde-C24108

Connection Oriented Architecture

StudentID	StudentName	Address
1	Aniket	Bhandup
2	Raj	Dadar
3	Tosif	Kalwa

PRACTICAL NO. 5: Webpage Demonstrating Disconnected Architecture
(ASP.NET Web Forms with SQL Server Database).

Webform.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication2.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <asp:Button ID="Button1" runat="server" OnClick="Button1_Click" Text="Button" />
            <br />
            <br />
            <asp:GridView ID="GridView1" runat="server">
            </asp:GridView>
        </div>
    </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Data.SqlClient;
using System.Data;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace PRAC_7_042
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            string connStr = "Data Source=NMITDLAB1-33\\SQLEXPRESS;Initial
Catalog=Sampledb;Integrated Security=True";
```

```
SqlDataAdapter da;
DataSet ds = new DataSet();
try
{
    using (SqlConnection conn = new SqlConnection(connStr))
    {
        string query = "Select * from Employee";
        da = new SqlDataAdapter(query, conn);
        da.Fill(ds, "Employee");
    }
    GridView1.DataSource = ds.Tables["Employee"];
    GridView1.DataBind();
}
catch (Exception ex)
{
    Response.Write("<script>alert('Error: " + ex.Message + "');</script>");
}
}
```

Output:

Aniket Shinde-C24108

Disconnected Architecture

StudentID	StudentName	Address
1	Aniket	Bhandup
2	Raj	Dadar
3	Tosif	Kalwa

PRACTICAL NO. 6: Create a webpage that demonstrates the use of data bound controls of ASP.NET.

Output:

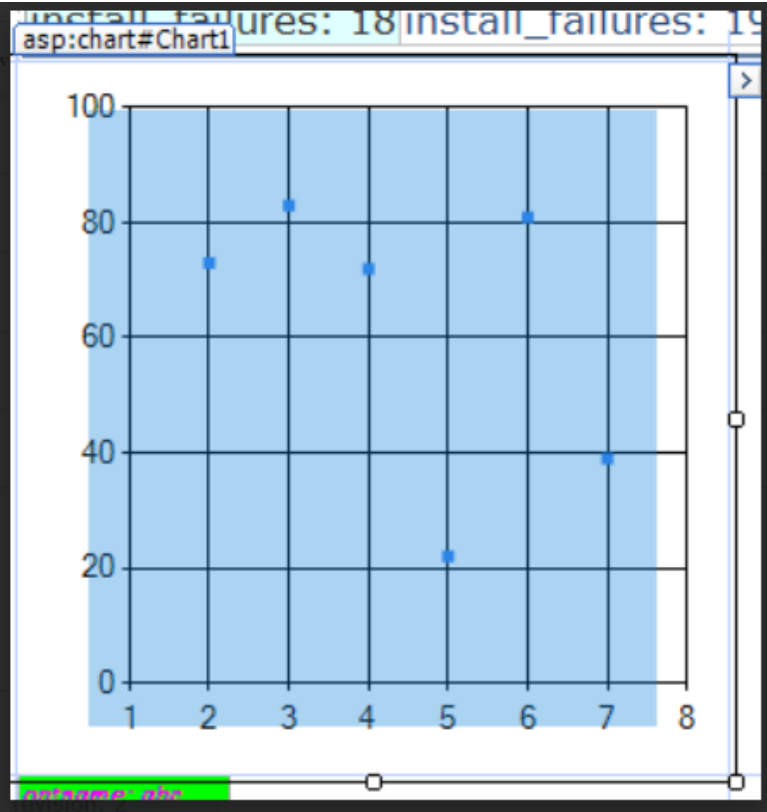
WebForm1.aspx

optname	value	major_version	minor_version	revision	install_failures
abc	☐	0	0	0	0
abc	☒	1	1	1	1
abc	☐	2	2	2	2
abc	☒	3	3	3	3
abc	☐	4	4	4	4

SqlDataSource1 - SqlDataSource1
asp:FormView#FormView1

optname: abc
value: ☐
major_version: 0
minor_version: 0
revision: 0
install_failures: 0

optname: abc value: ☐ major_version: 0 minor_version: 0 revision: 0 install_failures: 0	optname: abc value: ☐ major_version: 1 minor_version: 1 revision: 1 install_failures: 1	optname: abc value: ☐ major_version: 2 minor_version: 2 revision: 2 install_failures: 2
---	---	---



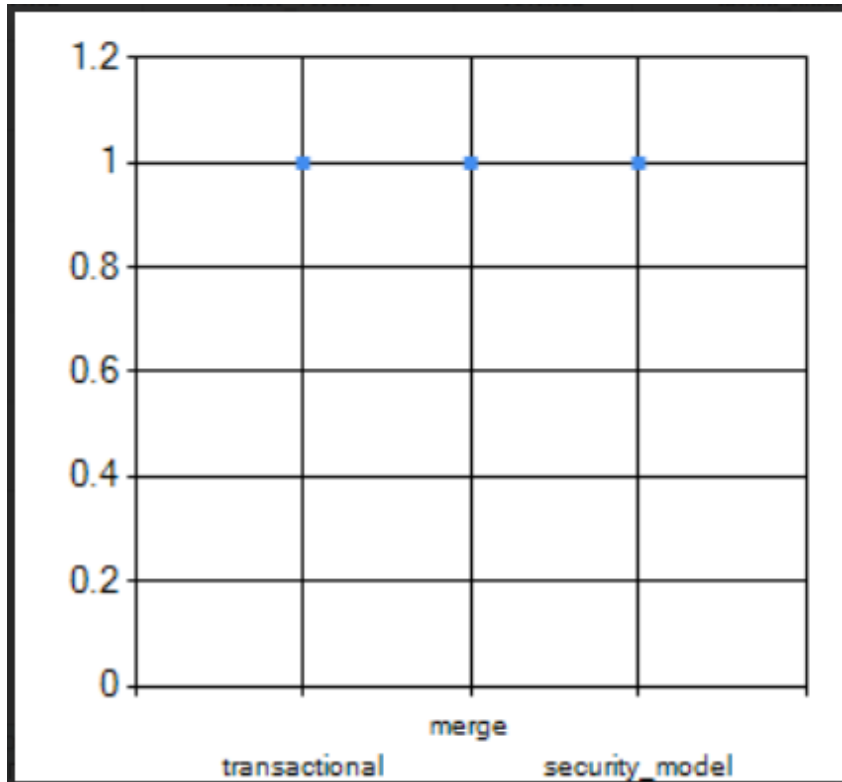
localhost:44342/WebForm1.aspx

optname	value	major_version	minor_version	revision	install_failures
transactional	☒	90	0	0	0
merge	☒	90	0	0	0
security_model	☒	90	0	0	0

optname: transactional
value: ☒
major_version: 90
minor_version: 0
revision: 0
install_failures: 0

optname: merge
value: ☒
major_version: 90
minor_version: 0
revision: 0
install_failures: 0

optname: security_model
value: ☒
major_version: 90
minor_version: 0
revision: 0
install_failures: 0



PRACTICAL NO. 7: Design a webpage to demonstrate the working of a simple stored procedure.**Webform.aspx:**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="practical_7.WebForm1" %>
<!DOCTYPE html>
<html>
<head>
    <title>Stored Procedure Demo</title>
</head>
<body>
    <form id="form1" runat="server">
        <h2>Stored Procedure Demonstration</h2>

        <asp:Button ID="btnGetUsers" runat="server" Text="Get Users"
OnClick="btnGetUsers_Click" />
        <br /><br />

        <asp:GridView ID="GridViewUsers" runat="server" AutoGenerateColumns="true"
BorderColor="Black" BorderWidth="1px" />
    </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Configuration;
using System.Data;
using System.Data.SqlClient;
using System.Web.UI;
namespace practical_7
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void btnGetUsers_Click(object sender, EventArgs e)
        {
            string connStr = "Data Source=DESKTOP-4EM5DMS\\SQLEXPRESS;Initial
Catalog=EmpDB;Integrated Security=True;Encrypt=False";
            using (SqlConnection conn = new SqlConnection(connStr))
            {
                using (SqlCommand cmd = new SqlCommand("GetUsers86", conn))
                {

```

```
cmd.CommandType = CommandType.StoredProcedure;
conn.Open();
SqlDataAdapter da = new SqlDataAdapter(cmd);
DataTable dt = new DataTable();
da.Fill(dt);
GridViewUsers.DataSource = dt;
GridViewUsers.DataBind();
    }
}
}
```

Output:

Stored Procedure Demonstration

Get Users

Id	Name	Email
1	Nishant	nshant@gmail.com
2	aniket	aniket@email.com
3	mansi	mansi@gmail.com
4	jayash	js@gmail.com
5	sid	s@gmail.com
6	anuj	a@gmail.com
7	sejal	sejal@gmail.com
8	sneha	sneha@gmail.com

DESKTOP-4EM5DMS\...B - dbo.EmpTable SQLQuery2.sql -			
	Id	Name	Email
	1	Nishant	nshant@gmail....
	2	Aniket	aniket.gmail.c...
	3	mansi	mansi@gmail.c...
	4	jayash	js@gmail.com
	5	sid	s@gmail.com
	6	anuj	a@gmail.com
	7	sejal	sejal@gmail.con
	8	sneha	sneha@gmail.c...
▶*	NULL	NULL	NULL

**PRACTICAL NO. 8: Design a webpage to demonstrate the working of
parameterized stored procedure.**

Webform.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication2.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <b>Parametrized stored procedure</b><br />
            <br />
            Enter Department Name:
            <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
            <br />
            <br />
            <asp:Button ID="Button1" runat="server" Text="Search" OnClick="Button1_Click"
/>
            <br />
            <br />
            <asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="False">
                <Columns>
                    <asp:BoundField DataField="cid" HeaderText="Customer ID" />
                    <asp:BoundField DataField="cname" HeaderText="Name" />
                    <asp:BoundField DataField="add" HeaderText="Address" />
                    <asp:BoundField DataField="salINR" HeaderText="Salary (INR)" />
                    <asp:BoundField DataField="salDollar" HeaderText="Salary (USD)" />
                </Columns>
            </asp:GridView>
        </div>
    </form>
</body>
</html>
```


Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Net.Mail;
using System.Data.SqlClient;
using System.Xml.Linq;
using System.Data;
namespace WebApplication2
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            string connStr = "Data Source=DESKTOP-TPEBNNC;Initial
Catalog=Student;Integrated Security=True";
            try
            {
                using (SqlConnection conn = new SqlConnection(connStr))
                {
                    using (SqlCommand cmd = new SqlCommand("Getdepartment", conn))
                    {
                        cmd.CommandType = CommandType.StoredProcedure;
                        cmd.Parameters.AddWithValue("@DepartmentName",
TextBox1.Text.Trim());
                        SqlDataAdapter da = new SqlDataAdapter(cmd);
                        DataTable dt = new DataTable();
                        da.Fill(dt);
                        GridView1.DataSource = dt;
                        GridView1.DataBind();
                    }
                }
            }
            catch (Exception ex)
            {
                Response.Write("<script>alert('An error occurred: " + ex.Message + "');</script>");
            }
        }
    }
}
```

Output:**Parametrized stored procedure**Enter Department Name:

Customer ID	Name	Address	Salary (INR)	Salary (USD)
1	Aniket	QWERTY	12345	5432

Parametrized stored procedureEnter Department Name:

Customer ID	Name	Address	Salary (INR)	Salary (USD)
4	Rajeev	CVBNM	567	234

```
/****** Script for SelectTopNRows command from SSMS *****/
SELECT TOP (1000) [cid]
      , [cname]
      , [add]
      , [salINR]
      , [salDollar]
FROM [Student].[dbo].[CustDW]
```

100 %

Results Messages

	cid	cname	add	salINR	salDollar
1	1	Nishant	QWERTY	12345	5432
2	2	Lucile	ASDFG	5432	234
3	3	Nathan	LKJHGF	12345	234
4	4	Rajeev	CVBNM	567	234
5	5	RDM	LKJHGFVB	7654	1234

SQLQuery4.sql - D:\...EBNNC\shaji (68)) SQLQuery2.sql - D:\...EBNNC\shaji (67))*

```
CREATE PROCEDURE Getdepartment
@DepartmentName nvarchar(50)
AS
BEGIN
Select * from CustDW WHERE cid = @DepartmentName;
END;
```

100 %

Messages

Commands completed successfully.

Completion time: 2025-05-20T16:07:45.8017581+05:30

PRACTICAL NO. 9: Design a webpage to display the use of LINQ.**Webform.aspx:**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication2.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <b>Parametrized stored procedure</b><br />
            <br />
            Enter Department Name:
            <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
            <br />
            <br />
            <asp:Button ID="Button1" runat="server" Text="Search" OnClick="Button1_Click"
/>
            <br />
            <br />
            <asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="False">
<Columns>
            <asp:BoundField DataField="cid" HeaderText="Customer ID" />
            <asp:BoundField DataField="cname" HeaderText="Name" />
            <asp:BoundField DataField="add" HeaderText="Address" />
            <asp:BoundField DataField="salINR" HeaderText="Salary (INR)" />
            <asp:BoundField DataField="salDollar" HeaderText="Salary (USD)" />
        </Columns>
    </asp:GridView>
        </div>
    </form>
</body>
</html>
```

Webform.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Net.Mail;
using System.Data.SqlClient;
using System.Xml.Linq;
using System.Data;
namespace WebApplication2
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string connStr = "Data Source=DESKTOP-TPEBNNC;Initial
Catalog=Student;Integrated Security=True";
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void Button1_Click(object sender, EventArgs e)
        {
            DataTable dt = new DataTable();
            using (SqlConnection conn = new SqlConnection(connStr))
            {
                using (SqlCommand cmd = new SqlCommand("SELECT * FROM CustDW", conn))
                {
                    SqlDataAdapter da = new SqlDataAdapter(cmd);
                    da.Fill(dt);
                }
            }
            var result = from row in dt.AsEnumerable()
                where row.Field<int>("salINR") > 10000
                select new
                {
                    CID = row.Field<int>("cid"),
                    Name = row.Field<string>("cname"),
                    Address = row.Field<string>("add"),
                    SalaryINR = row.Field<int>("salINR"),
                    SalaryUSD = row.Field<int>("salDollar")
                };
            GridView1.DataSource = result.ToList();
            GridView1.DataBind();
        }
    }
}
```

Output:**LINQ APPLICATION**

Fetch Data

Customer ID	Name	Address	Salary (INR)	Salary (USD)
1	Aniket	QWERTY	12345	5432
3	Nathan	LKJHGF	12345	234

```
/****** Script for SelectTopNRows command from SSMS *****/  
SELECT TOP (1000) [cid]  
    , [cname]  
    , [add]  
    , [salINR]  
    , [salDollar]  
FROM [Student].[dbo].[CustDW]
```

100 %

Results Messages

	cid	cname	add	salINR	salDollar
1	1	Aniket	QWERTY	12345	5432
2	2	Lucile	ASDFG	5432	234
3	3	Nathan	LKJHGF	12345	234
4	4	Rajeev	CVBNM	567	234
5	5	RDM	LKJHGFVB	7654	1234

PRACTICAL NO. 10: Build websites to demonstrate the working of entity frameworks in dot net.

1 SQL Server: Create a Database & Table

Open **SQL Server Management Studio (SSMS)** and execute:

```
CREATE DATABASE EmployeeDB;  
USE EmployeeDB;
```

```
CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY IDENTITY(1,1),  
    Name NVARCHAR(100),  
    Department NVARCHAR(100),  
    Salary DECIMAL(10,2)  
);
```

```
INSERT INTO Employees (Name, Department, Salary)  
VALUES ('John Doe', 'IT', 60000), ('Jane Smith', 'HR', 55000);
```

2 Create an ASP.NET Web Application

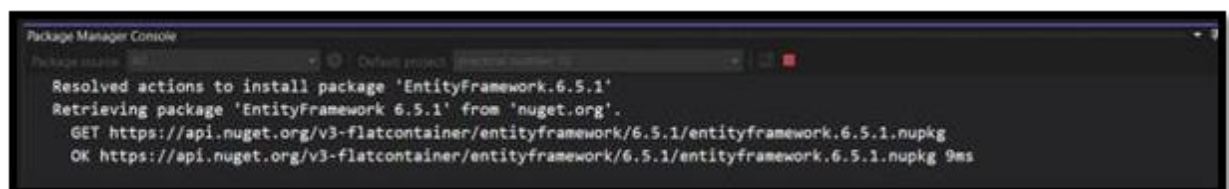
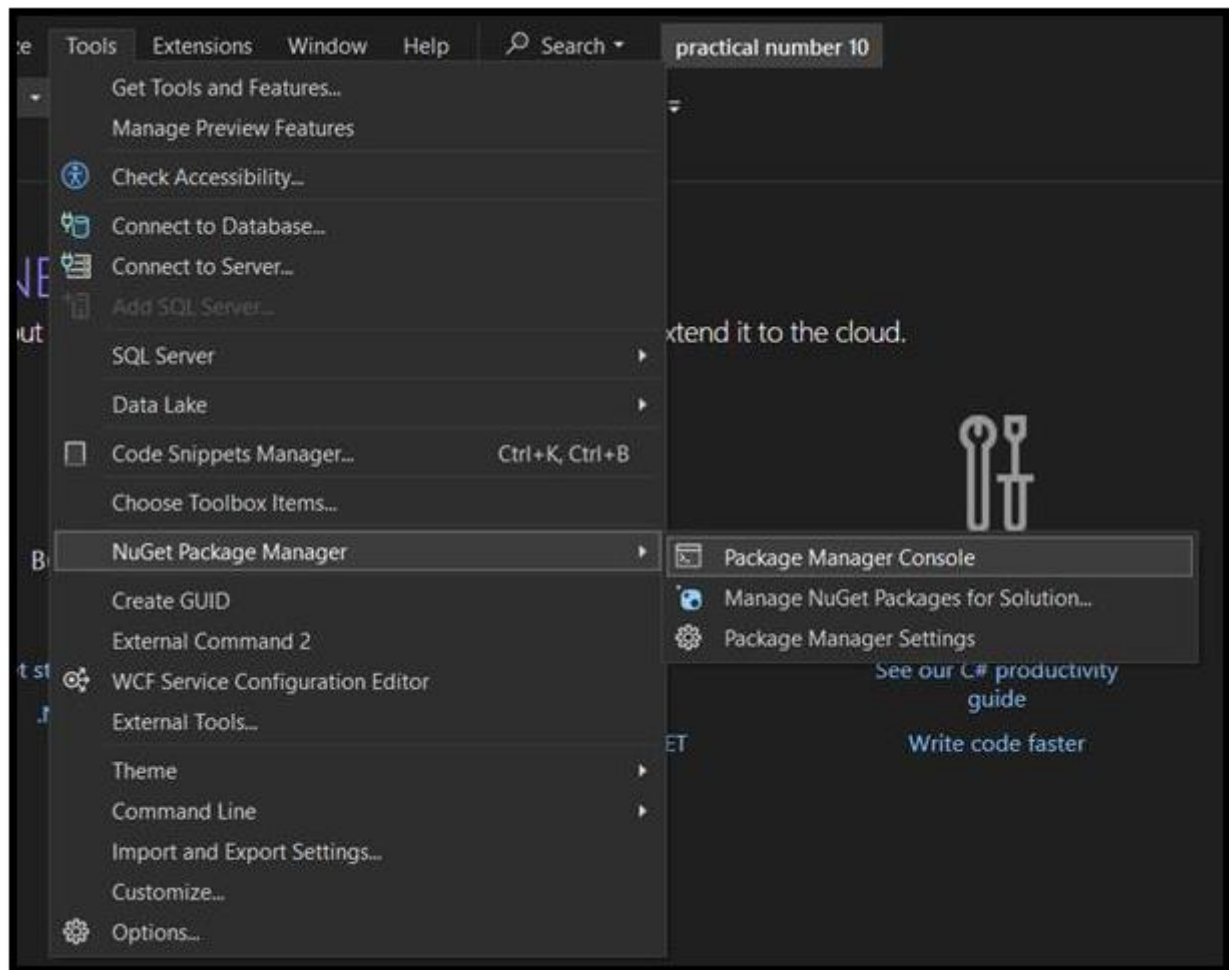
- Open Visual Studio → Create a New Project → Choose ASP.NET Web Application (.NET Framework)
- Select Web Forms and click Create

3 Install Entity Framework

Open **Package Manager Console** (Tools → NuGet Package Manager → Package Manager Console)

Run the command:

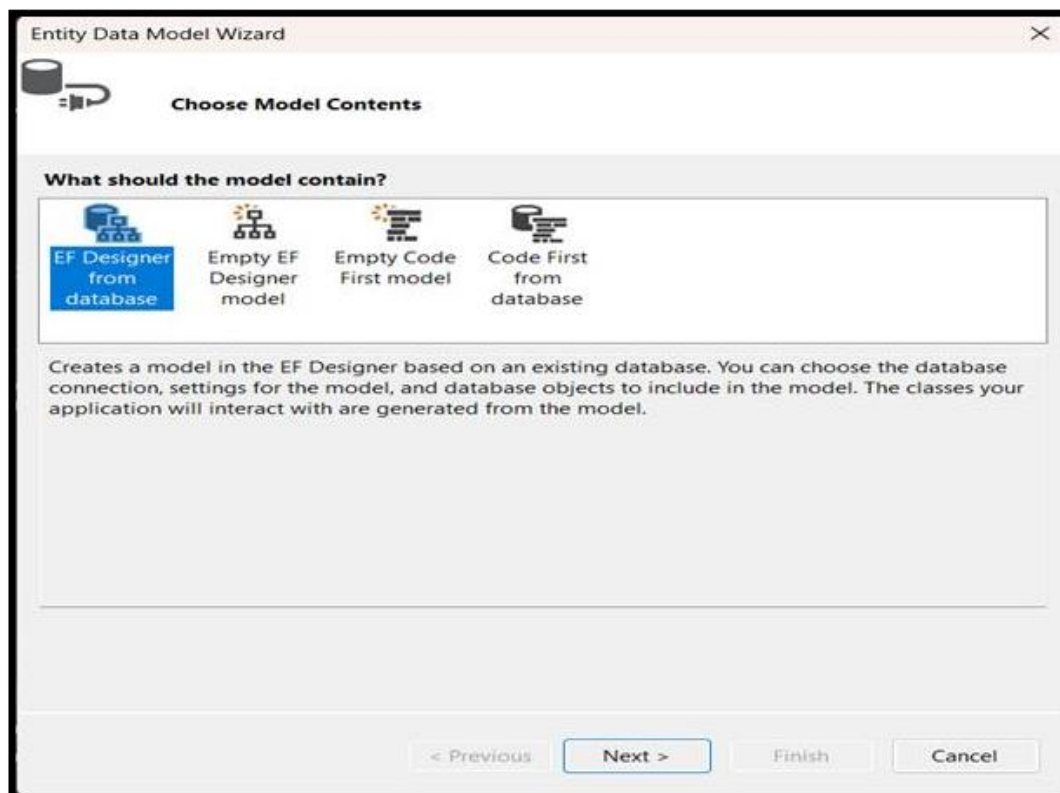
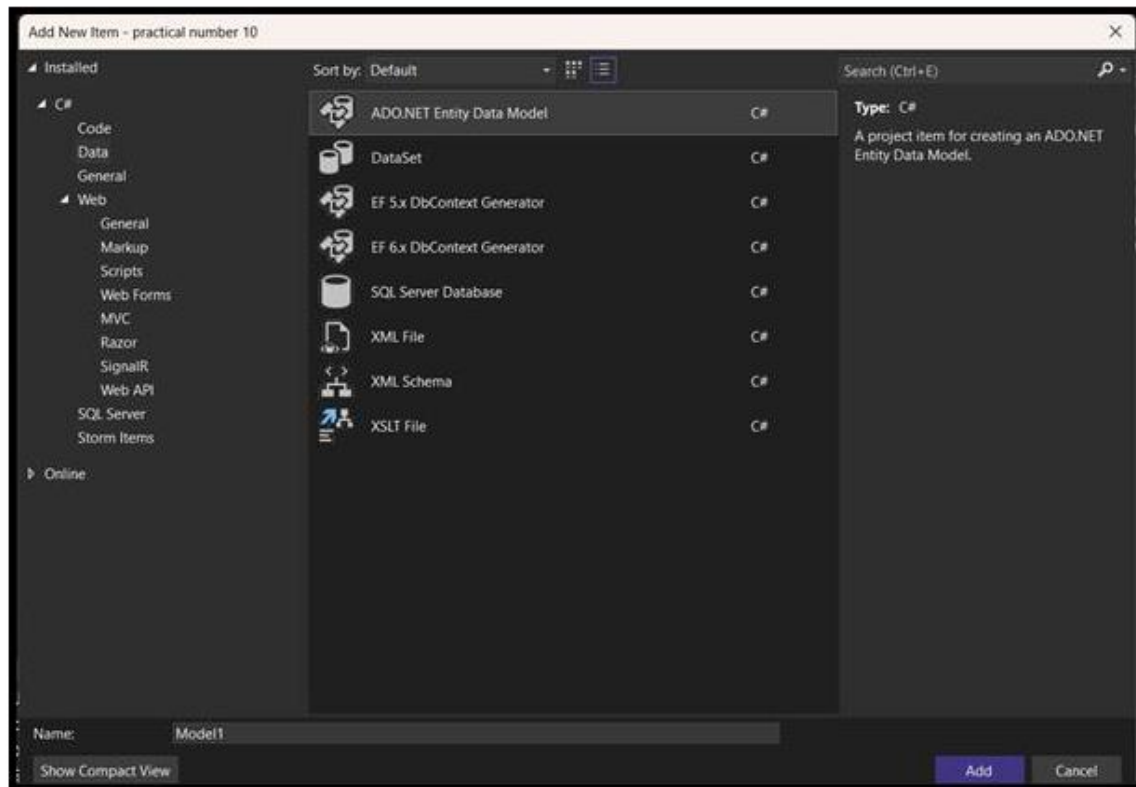
Install-Package EntityFramework



4 Create Entity Framework Model

1. Right-click the project → Add → New Item → Select **ADO.NET Entity Data Model**
2. Choose **EF Designer from Database**
3. Select "**EmployeeDB**" as the database
4. Select the **Employees** table → Finish

This generates the **Employee.cs** model class.



Entity Data Model Wizard



Choose Your Data Connection

Which data connection should your application use to connect to the database?

laptop-v83lpsdi\sqlexpress.EmployeeDB.dbo

New Connection...

This connection string appears to contain sensitive data (for example, a password) that is required to connect to the database. Storing sensitive data in the connection string can be a security risk. Do you want to include this sensitive data in the connection string?

☐ No, exclude sensitive data from the connection string. I will set it in my application code.

☐ Yes, include the sensitive data in the connection string.

Connection string:

metadata=res://*/Employee.csdl|res://*/Employee.ssdl|
res://*/Employee.msl;provider=System.Data.SqlClient;provider connection string="data
source=LAPTOP-V83LPSDI\SQLEXPRESS;initial catalog=EmployeeDB;integrated
security=True;trustservercertificate=True;MultipleActiveResultSets=True;App=EntityFramework"

☒ Save connection settings in Web.Config as:

EmployeeDBEntities

< Previous

Next >

Finish

Cancel

Entity Data Model Wizard

Choose Your Database Objects and Settings

Which database objects do you want to include in your model?

- ☒ Tables
 - ☒ dbo
 - ☐ Views
 - ☐ Stored Procedures and Functions

☒ Pluralize or singularize generated object names

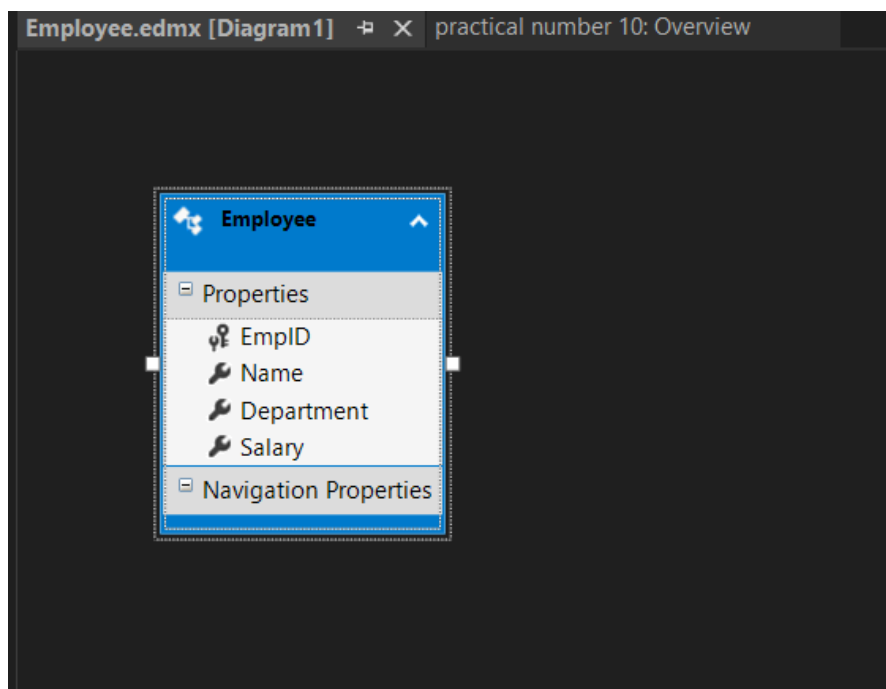
☒ Include foreign key columns in the model

☒ Import selected stored procedures and functions into the entity model

Model Namespace:

EmployeeDBModel

< Previous Next > Finish Cancel



5 Create an ASP.NET Web Form (WebForm1.aspx)

Modify `WebForm1.aspx` to include a **GridView** to display records and a **Form** for adding new employees.

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="practical_number_10.WebForm1" %>

<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <title>Entity Framework CRUD Demo</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <h2>Employee Management (Entity Framework)</h2>
            <asp:Label runat="server" Text="Name:"></asp:Label>
            <asp:TextBox ID="txtName" runat="server" style="margin-left: 46px"></asp:TextBox>
            <br />
            <asp:Label runat="server" Text="Department:"></asp:Label>
            <asp:TextBox ID="txtDepartment" runat="server"></asp:TextBox> <br />
            <asp:Label runat="server" Text="Salary:"></asp:Label>
            <asp:TextBox ID="txtSalary" runat="server" style="margin-left: 41px"></asp:TextBox>
            <br />      <br />
            <br />
            <asp:Button ID="btnAdd" runat="server" Text="Add Employee" OnClick="btnAdd_Click" />
            <br /><br />
            <!-- Display Employees -->
            <asp:GridView ID="gvEmployees" runat="server" AutoGenerateColumns="False"
DataKeyNames="EmpID"
    OnRowEditing="gvEmployees_RowEditing" OnRowUpdating="gvEmployees_RowUpdating"
    OnRowCancelingEdit="gvEmployees_RowCancelingEdit"
OnRowDeleting="gvEmployees_RowDeleting" Height="233px" Width="677px">
                <Columns>
                    <asp:BoundField DataField="EmpID" HeaderText="EmpID" ReadOnly="True" />
                    <asp:BoundField DataField="Name" HeaderText="Name" />
                    <asp:BoundField DataField="Department" HeaderText="Department" />
                    <asp:BoundField DataField="Salary" HeaderText="Salary" />
                    <asp:CommandField ShowEditButton="True" ShowDeleteButton="True" />
                </Columns>
            </asp:GridView>
        </div>
    </form>
</body>
</html>
```

Employee Management (Entity Framework)

Name:

Department:

Salary:

EmpID	Name	Department	Salary	
Databound	Databound	Databound	Databound	Edit Delete
Databound	Databound	Databound	Databound	Edit Delete
Databound	Databound	Databound	Databound	Edit Delete
Databound	Databound	Databound	Databound	Edit Delete
Databound	Databound	Databound	Databound	Edit Delete

6 Backend Code (WebForm1.aspx.cs)

Modify WebForm1.aspx.cs to implement CRUD operations using **Entity Framework**.

```
using System;
using System.Linq;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace practical_number_10
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        EmployeeDBEntities1 db = new EmployeeDBEntities1 ();
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                LoadEmployees();
            }
        }
        private void LoadEmployees()
        {
            gvEmployees.DataSource = db.Employees.ToList();
            gvEmployees.DataBind();
        }
        protected void btnAdd_Click(object sender, EventArgs e)
        {

```

```
Employee emp = new Employee
{
    Name = txtName.Text,
    Department = txtDepartment.Text,
    Salary = Convert.ToDecimal(txtSalary.Text)
};
db.Employees.Add(emp);
db.SaveChanges();
LoadEmployees();
}

protected void gvEmployees_RowEditing(object sender, GridViewEditEventArgs e)
{
    gvEmployees.EditIndex = e.NewEditIndex;
    LoadEmployees();
}

protected void gvEmployees_RowUpdating(object sender, GridViewUpdateEventArgs e)
{
    int empID = Convert.ToInt32(gvEmployees.DataKeys[e.RowIndex].Value);
    Employee emp = db.Employees.Find(empID);
    TextBox txtName =
(TextBox)gvEmployees.Rows[e.RowIndex].Cells[1].Controls[0];
    TextBox txtDepartment =
(TextBox)gvEmployees.Rows[e.RowIndex].Cells[2].Controls[0];
    TextBox txtSalary =
(TextBox)gvEmployees.Rows[e.RowIndex].Cells[3].Controls[0];
    emp.Name = txtName.Text;
    emp.Department = txtDepartment.Text;
    emp.Salary = Convert.ToDecimal(txtSalary.Text);
    db.SaveChanges();
    gvEmployees.EditIndex = -1;
    LoadEmployees();
}

protected void gvEmployees_RowCancelingEdit(object sender,
GridViewCancelEventArgs e)
{
    gvEmployees.EditIndex = -1;
    LoadEmployees();
}

protected void gvEmployees_RowDeleting(object sender, GridViewDeleteEventArgs e)
{
    int empID = Convert.ToInt32(gvEmployees.DataKeys[e.RowIndex].Value);
    Employee emp = db.Employees.Find(empID);
    db.Employees.Remove(emp);
    db.SaveChanges();
    LoadEmployees();
}

}
}
```

Entity Framework CRUD Demo

localhost:44356/WebForm1.aspx

New TabOBE | MasterSoftAdobe Acrobat

Employee Management (Entity Framework)

Name:

Department:

Salary:

Add Employee

EmpID	Name	Department	Salary	
1	John Doe	IT222	60000.00	Edit Delete
2	Jane Smith	HR	55000.00	Edit Delete

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_Number_11
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (ViewState["UserName"] != null)
            {

```



```
        lblMessage.Text = "Stored in View State: " + ViewState["UserName"].ToString();
    }

}

protected void btnSubmit_Click(object sender, EventArgs e)
{
    ViewState["UserName"] = txtName.Text;
    lblMessage.Text = "Data Saved in View State!";
}

}
}
```

Data Saved in View State! Aniket Shinde

2 Query String (Passing Data in URL): Query strings pass data between pages using the URL.

WebForm2.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="Practical_Number_11.WebForm2" %>

<!DOCTYPE html>
<html>
<head>
    <title>Query String Example</title>
</head>
<body>
    <form runat="server">
        <asp:TextBox ID="txtName" runat="server" Height="67px" Width="193px"></asp:TextBox>
        <br />
        <br />
        <br />
        <asp:Button ID="btnSubmit" runat="server" Text="Go to Next Page" OnClick="btnSubmit_Click"
Height="62px" Width="203px" />
    </form>
</body>
</html>
```

WebForm2.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_Number_11
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void btnSubmit_Click(object sender, EventArgs e)
        {
            Response.Redirect("WebForm3.aspx?name=" + txtName.Text);
        }
    }
}
```

}

Aniket Shinde

Go to Next Page

WebForm3.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm3.aspx.cs"
Inherits="Practical_Number_11.WebForm3" %>
<!DOCTYPE html>
<html>
<head>
  <title>Query String Result</title>
</head>
<body>
  <form runat="server">
    <asp:Label ID="lblMessage" runat="server"></asp:Label>
  </form>
</body>
</html>
```

WebForm3.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_Number_11
{
    public partial class WebForm3 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Request.QueryString["name"] != null)
            {
                lblMessage.Text = "Query String Value: " + Request.QueryString["name"];
            }
        }
    }
}
```

Query String Value: Aniket Shinde

3 Cookies (Persistent Client-Side Storage): Stores small data on the client-side, accessible across pages.

WebForm4.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm4.aspx.cs" Inherits="Practical_Number_11.WebForm4" %>

<!DOCTYPE html>
<html>
<head>
    <title>Cookies Example</title>
</head>
<body>
    <form runat="server">
        <asp:TextBox ID="txtName" runat="server" Height="55px" Width="226px"></asp:TextBox>
        <br />
        <br />
        <asp:Button ID="btnSetCookie" runat="server" Text="Set Cookie" OnClick="btnSetCookie_Click"
Height="56px" style="margin-left: 34px" Width="143px" />
        <br />
        <br />
        <asp:Button ID="btnGetCookie" runat="server" Text="Get Cookie"
OnClick="btnGetCookie_Click" Height="51px" style="margin-left: 40px" Width="138px" />
        <br />
        <br />
        &nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&~
        <asp:Label ID="lblMessage" runat="server"></asp:Label>
    </form>
</body>
</html>
```

WebForm4.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_Number_11
{
    public partial class WebForm4 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void btnSetCookie_Click(object sender, EventArgs e)
        {
            HttpCookie cookie = new HttpCookie("UserName", txtName.Text);
        }
    }
}
```

```
        cookie.Expires = DateTime.Now.AddDays(7); // Cookie valid for 7 days
        Response.Cookies.Add(cookie);
        lblMessage.Text = "Cookie set successfully!";
    }
    protected void btnGetCookie_Click(object sender, EventArgs e)
    {
        HttpCookie cookie = Request.Cookies["UserName"];
        if (cookie != null)
        {
            lblMessage.Text = "Stored Cookie Value: " + cookie.Value;
        }
        else
        {
            lblMessage.Text = "No Cookie Found!";
        }
    }
}
```

Cookie Set Succesfully !

Stored Cookie Value : Aniket Shinde

4 Hidden Fields (Storing Temporary Data): Stores data in an HTML form but remains invisible to the user.

WebForm5.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm5.aspx.cs"
Inherits="Practical_Number_11.WebForm5" %>
<html>
<head>
    <title>Hidden Field Example</title>
</head>
<body>
    <form runat="server">
        <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
        <asp:HiddenField ID="hiddenField" runat="server" Value="12345" />
        <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
        <asp:Label ID="lblMessage" runat="server"></asp:Label>
    </form>
</body>
</html>
```

WebForm5.aspx.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_Number_11
{
    public partial class WebForm5 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }
        protected void btnSubmit_Click(object sender, EventArgs e)
        {
            lblMessage.Text = "Entered Name: " + txtName.Text + ", Hidden ID: " + hiddenField.Value;
        }
    }
}
```


PRACTICAL NO. 12: Design Web Applications using Server-Side Session Management Techniques

1. Create an ASP.NET Web Forms Project

1. Open **Visual Studio**.
2. Click **Create a new project**.
3. Select **ASP.NET Web Application (.NET Framework)** and click **Next**.
4. Name the project (e.g., SessionManagementDemo).
5. Choose **Web Forms** and click **Create**.

2. Configure Session Management in Web.config

```
<configuration>
  <system.web>
    <compilation debug="true" targetFramework="4.7.2" />
    <httpRuntime targetFramework="4.7.2" />
    <sessionState mode="InProc" timeout="20" cookieless="UseCookies" />
  </system.web>
```

Modify Web.config to use **InProc (default)**, **StateServer**, or **SQL Server**.

For **InProc Session** (default):

```
<sessionState mode="InProc" timeout="20" cookieless="UseCookies" />
```

For **StateServer Session**:

```
<sessionState mode="StateServer" stateConnectionString="tcpip=127.0.0.1:42424"
timeout="20"/>
```

For **SQL Server Session**:

```
<sessionState mode="SQLServer" sqlConnectionString="data source=SQLSERVER;Initial
Catalog=SessionDB;Integrated Security=True" timeout="20"/>
```

3. Implement Login Page (Login.aspx)

Frontend: Login.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Login.aspx.cs"
Inherits="SessionManagementDemo.Login" %>

<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <title>Login</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <h2>Login Page</h2>
            <asp:Label ID="lblMessage" runat="server" ForeColor="Red"></asp:Label>
            <br />
            Username: <asp:TextBox ID="txtUsername" runat="server"></asp:TextBox>
            <br />
            Password: <asp:TextBox ID="txtPassword" runat="server" TextMode="Password"></asp:TextBox>
            <br />
            <asp:Button ID="btnLogin" runat="server" Text="Login" OnClick="btnLogin_Click" />
        </div>
    </form>
</body>
</html>
```

Backend: Login.aspx.cs

```
using System;
using System.Web;
namespace SessionManagementDemo
{
    public partial class Login : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Session["Username"] != null)
            {
                Response.Redirect("Dashboard.aspx");
            }
        }
        protected void btnLogin_Click(object sender, EventArgs e)
        {
            string username = txtUsername.Text;
            string password = txtPassword.Text;
            if (username == "admin" && password == "1234")
            {
                Session["Username"] = username;
                Response.Redirect("Dashboard.aspx");
            }
            else
            {
                lblMessage.Text = "Invalid username or password!";
            }
        }
    }
}
```

4. Create a Dashboard Page (Dashboard.aspx)

Frontend: Dashboard.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Dashboard.aspx.cs"
Inherits="SessionManagementDemo.Dashboard" %>

<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <title>Dashboard</title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
            <h2>Welcome, <asp:Label ID="lblUser" runat="server"></asp:Label>!</h2>
            <asp:Button ID="btnLogout" runat="server" Text="Logout" OnClick="btnLogout_Click" />
        </div>
    </form>
</body>
</html>
```

Backend: Dashboard.aspx.cs

```
using System;
using System.Web;

namespace SessionManagementDemo
{
    public partial class Dashboard : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Session["Username"] == null)
            {
                Response.Redirect("Login.aspx");
            }
            else
            {
                lblUser.Text = Session["Username"].ToString();
            }
        }

        protected void btnLogout_Click(object sender, EventArgs e)
        {
            Session.Abandon();
            Response.Redirect("Login.aspx");
        }
    }
}
```

Login Page

Username:	Aniket
Password:	
Login	

Welcome, Aniket

Logout

PRACTICAL NO. 13: Build a web page using AJAX Controls.

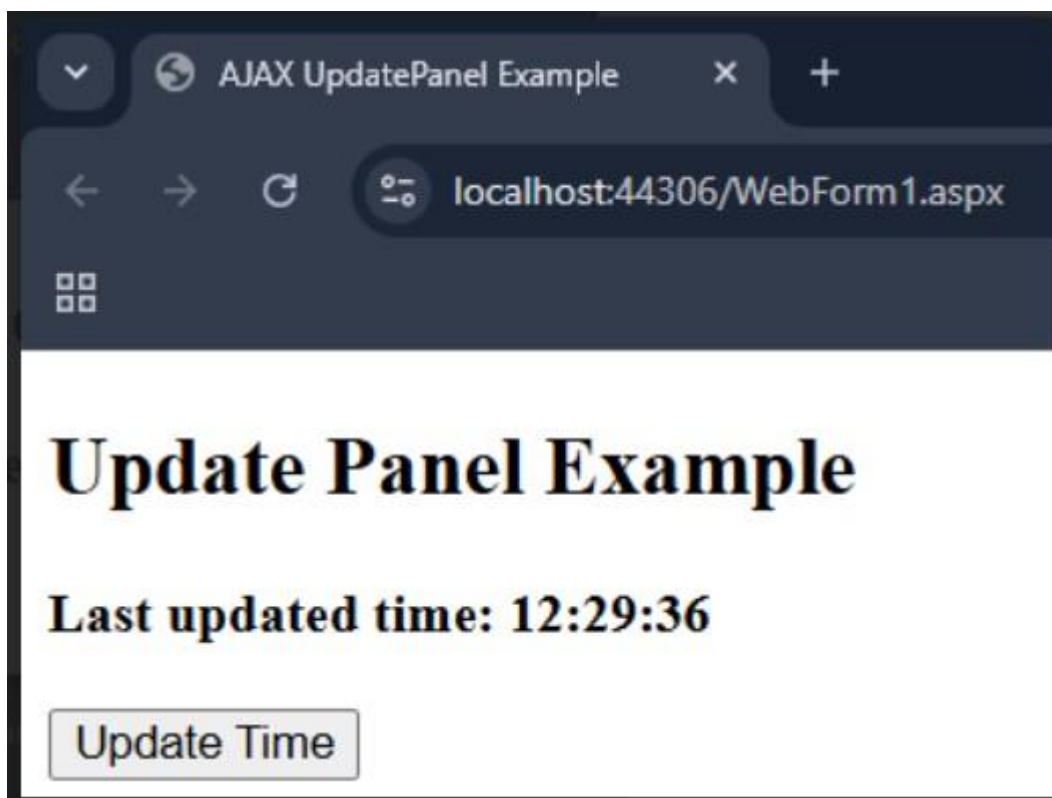
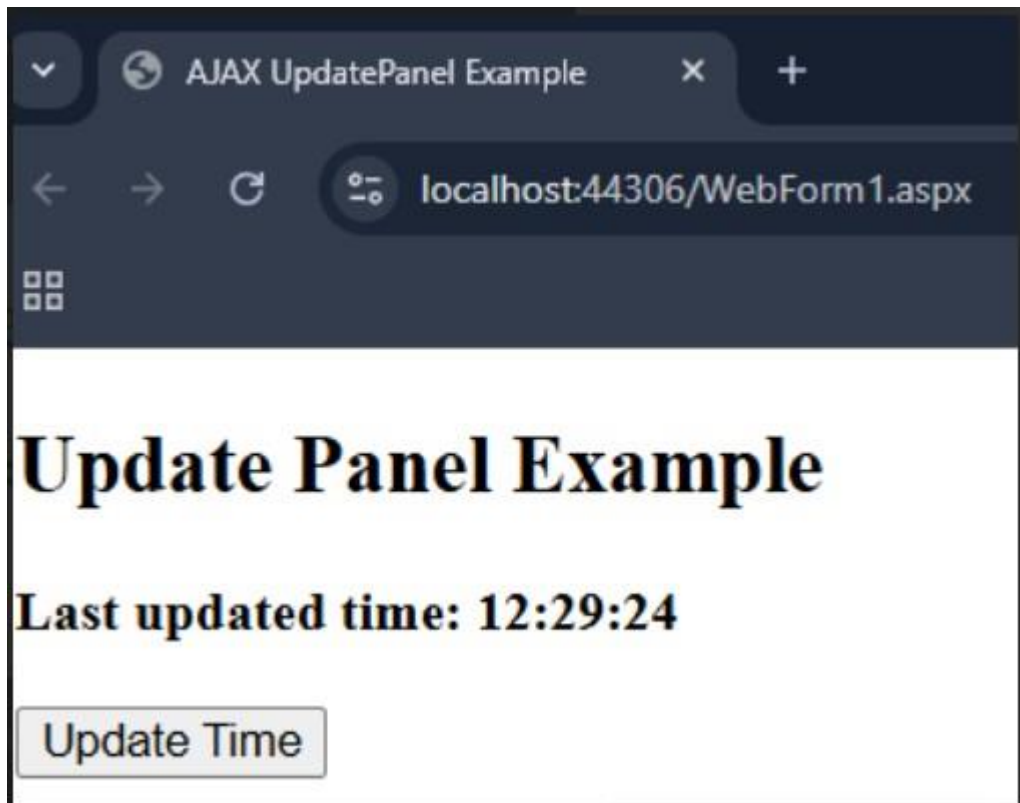
1st Program: AJAX UpdatePanel Example (Partial Page Update): This program updates the **current time** without refreshing the entire page.

Frontend: WebForm1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="AjaxDemo.WebForm1" %>
<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <title>AJAX UpdatePanel Example</title>
</head>
<body>
    <form id="form1" runat="server">
        <asp:ScriptManager ID="ScriptManager1" runat="server"></asp:ScriptManager>
        <h2>Update Panel Example</h2>
        <asp:UpdatePanel ID="UpdatePanel1" runat="server">
            <ContentTemplate>
                <asp:Label ID="lblTime" runat="server" Font-Bold="True"></asp:Label>
                <br /><br />
            </ContentTemplate>
        </asp:UpdatePanel>
        <asp:Button ID="btnUpdate" runat="server" Text="Update Time" OnClick="btnUpdate_Click" />
    </form>
</body>
</html>
```

Backend: WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace AjaxDemo
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                lblTime.Text = "Last updated time: " + DateTime.Now.ToString("HH:mm:ss");
            }
        }
        protected void btnUpdate_Click(object sender, EventArgs e)
        {
            lblTime.Text = "Last updated time: " + DateTime.Now.ToString("HH:mm:ss");
        }
    }
}
```



2nd Program: AJAX Timer Example (Auto Refresh without Full Page Load):

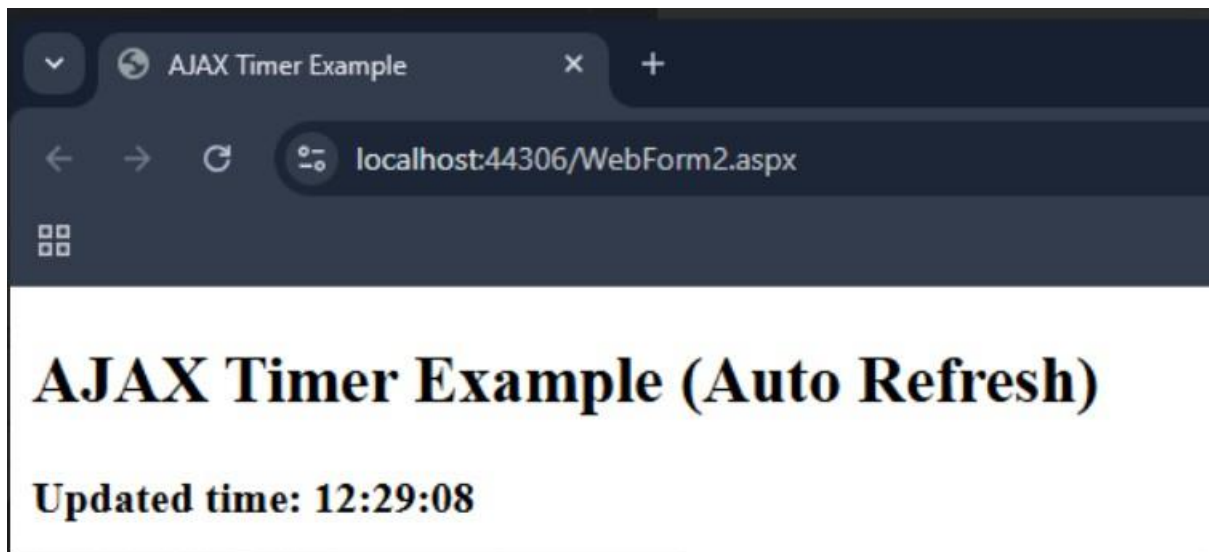
This program **automatically updates the time every 5 seconds** using an **AJAX Timer Control**.

Frontend: WebForm2.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"
Inherits="AjaxDemo.WebForm2" %>
<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <title>AJAX Timer Example</title>
</head>
<body>
    <form id="form1" runat="server">
        <asp:ScriptManager ID="ScriptManager1" runat="server"></asp:ScriptManager>
        <h2>AJAX Timer Example (Auto Refresh)</h2>
        <asp:UpdatePanel ID="UpdatePanel1" runat="server">
            <ContentTemplate>
                <asp:Label ID="lblTime" runat="server" Font-Bold="True"></asp:Label>
            </ContentTemplate>
        </asp:UpdatePanel>
        <asp:Timer ID="Timer1" runat="server" Interval="5000" OnTick="Timer1_Tick" />
    </form>
</body>
</html>
```

Backend: TimerDemo.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace AjaxDemo
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                lblTime.Text = "Current time: " + DateTime.Now.ToString("HH:mm:ss");
            }
        }
        protected void Timer1_Tick(object sender, EventArgs e)
        {
            lblTime.Text = "Updated time: " + DateTime.Now.ToString("HH:mm:ss");
        }
    }
}
```



PRACTICAL NO. 14: Build a web application to create and use web service in ASP.net.

1. Create New ASP.NET Web Forms App

- Open Visual Studio
- Create a new project:
ASP.NET Web Application (.NET Framework)
 - Name it: CalculatorApp
 - Template: **Empty**
 - Check: **Web Forms**
 - Click **Create**

2. Add ASMX Web Service

- Right-click project → **Add** → **New Item**
- Select **Web Service (ASMX)**
- Name it: CalculatorService.asmx

CalculatorService.asmx.cs

using System.Web.Services;

```
namespace CalculatorApp
{
    /// <summary>
    /// Web service for basic calculator operations
    /// </summary>
    [WebService(Namespace = "http://tempuri.org/")]
    [WebServiceBinding(ConformsTo = WsiProfiles.BasicProfile1_1)]
    public class CalculatorService : WebService
    {
        [WebMethod]
        public int Add(int a, int b)
        {
            return a + b;
        }

        [WebMethod]
        public int Subtract(int a, int b)
        {
            return a - b;
        }
    }
}
```

3. Add a Web Form

- Right-click project → **Add** → **New Item**
- Choose **Web Form** → Name: Default.aspx

Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="CalculatorApp.Default" %>

<!DOCTYPE html>
<html>
<head runat="server">
  <title>Calculator Web Service Client</title>
</head>
<body>
  <form id="form1" runat="server">
    <div>
      Number 1: <asp:TextBox ID="txtA" runat="server" /><br />
      Number 2: <asp:TextBox ID="txtB" runat="server" /><br /><br />
      <asp:Button ID="btnAdd" runat="server" Text="Add" OnClick="btnAdd_Click" />
      <asp:Button ID="btnSub" runat="server" Text="Subtract" OnClick="btnSub_Click"
/><br /><br />
      Result: <asp:Label ID="lblResult" runat="server" Text="" />
    </div>
  </form>
</body>
</html>
```

4. Add the Service Reference (Self Reference)

- Build the project first (Ctrl+Shift+B)
- Right-click the project → **Add Service Reference**
- Address: `http://localhost:<your-port>/CalculatorService.asmx`
 - (Run the project and copy the URL from the browser)
- Namespace: CalcRef
- Click **OK**

5. Code-Behind for Default.aspx.cs

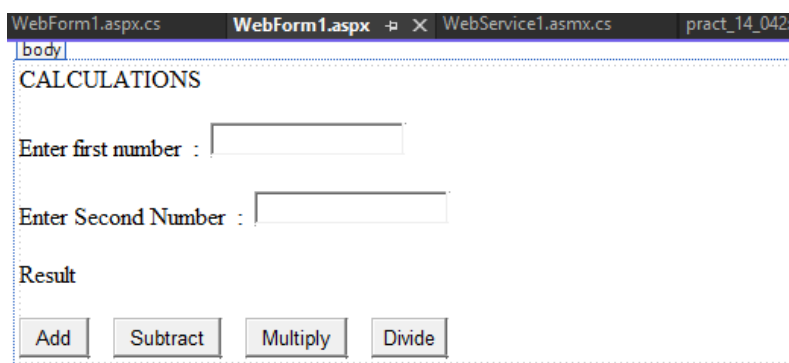
```
using System;
using CalculatorApp.CalcRef;

namespace CalculatorApp
{
    public partial class Default : System.Web.UI.Page
    {
        CalculatorServiceSoapClient client;

        protected void Page_Load(object sender, EventArgs e)
        {
            client = new CalculatorServiceSoapClient();
        }

        protected void btnAdd_Click(object sender, EventArgs e)
        {
            int a = int.Parse(txtA.Text);
            int b = int.Parse(txtB.Text);
            int result = client.Add(a, b);
            lblResult.Text = "Result: " + result;
        }

        protected void btnSub_Click(object sender, EventArgs e)
        {
            int a = int.Parse(txtA.Text);
            int b = int.Parse(txtB.Text);
            int result = client.Subtract(a, b);
            lblResult.Text = "Result: " + result;
        }
    }
}
```



The screenshot shows a web browser window with the title bar displaying 'WebForm1.aspx.cs', 'WebForm1.aspx', 'WebService1.asmx.cs', and 'pract_14_042:'. The main content area is titled 'CALCULATIONS' and contains two input fields for numbers. The first input field is preceded by the text 'Enter first number :'. The second input field is preceded by the text 'Enter Second Number :'. Below the input fields, the word 'Result' is displayed. At the bottom of the form, there are four buttons labeled 'Add', 'Subtract', 'Multiply', and 'Divide'.

← ↻ <https://localhost:44372/WebForm1.aspx>

CALCULATIONS

Enter first number :

Enter Second Number :

Addition = 15

← ↻ <https://localhost:44372/WebForm1.aspx>

CALCULATIONS

Enter first number :

Enter Second Number :

Subtraction = 5

← ↻ <https://localhost:44372/WebForm1.aspx>

CALCULATIONS

Enter first number :

Enter Second Number :

Multiplication = 50

← ↻ <https://localhost:44372/WebForm1.aspx>

CALCULATIONS

Enter first number :

Enter Second Number :

Division = 2

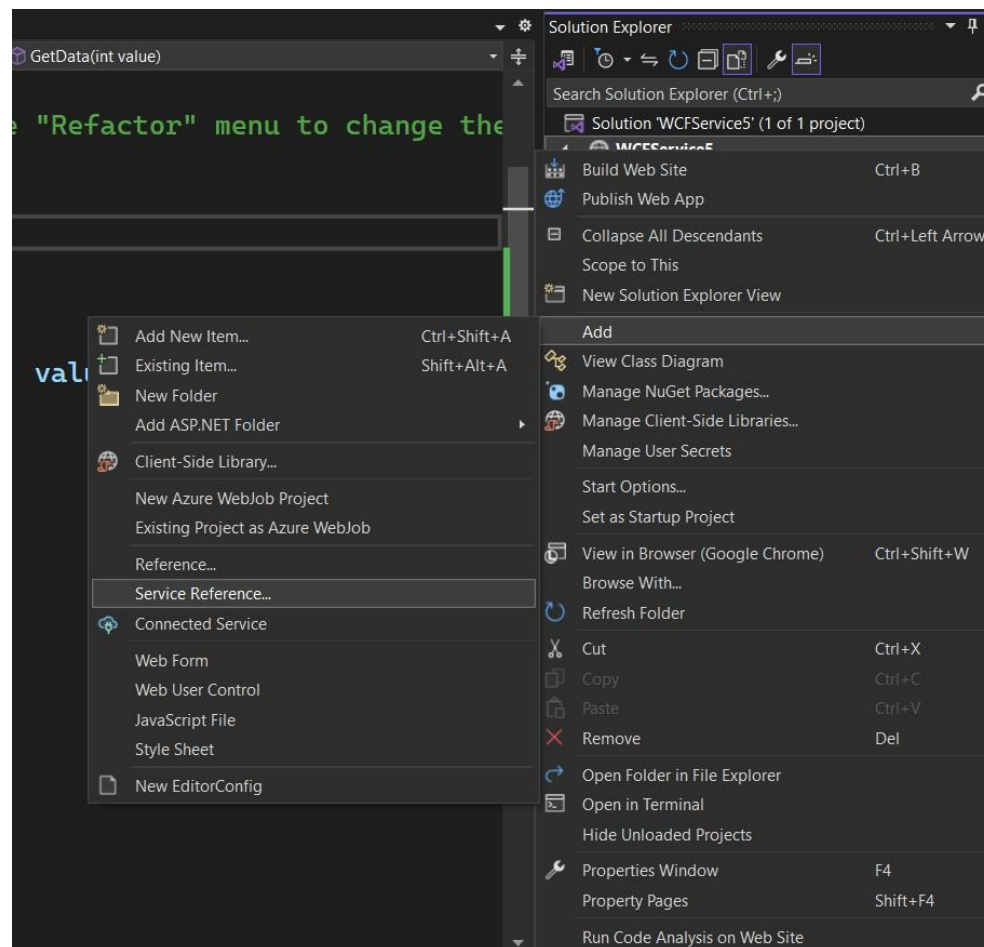
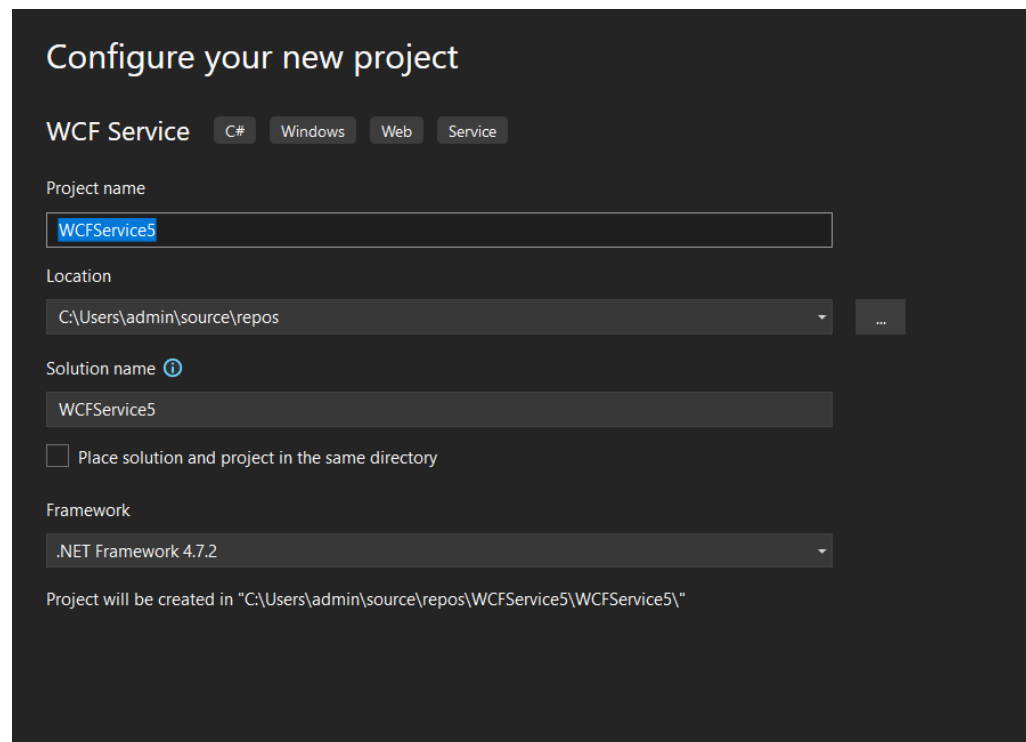
PRACTICAL NO. 15: Build a web application to create and WCF service in ASP.net.

Iservice.cs :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Runtime.Serialization;
using System.ServiceModel;
using System.ServiceModel.Web;
using System.Text;
// NOTE: You can use the "Rename" command on the "Refactor" menu to change the
interface name "IService" in both code and config file together.
[ServiceContract]
public interface IService
{
    [OperationContract]
    string GetData(int value);
    [OperationContract]
    double add(double a, double b);
    [OperationContract]
    double sub(double a, double b);
    [OperationContract]
    double mul(double a, double b);
    [OperationContract]
    double div(double a, double b);
    [OperationContract]
    CompositeType GetDataUsingDataContract(CompositeType composite);
}
[DataContract]
public class CompositeType
{
    bool boolValue = true;
    string stringValue = "Hello ";
    [DataMember]
    public bool BoolValue
    {
        get { return boolValue; }
        set { boolValue = value; }
    }
    [DataMember]
    public string StringValue
    {
        get { return stringValue; }
        set { stringValue = value; }
    }
}
```

Services.cs :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Runtime.Serialization;
using System.ServiceModel;
using System.ServiceModel.Web;
using System.Text;
// NOTE: You can use the "Rename" command on the "Refactor" menu to change the class
name "Service" in code, svc and config file together.
public class Service : IService
{
    public string GetData(int value)
    {
        return string.Format("You entered: {0}", value);
    }
    public double add(double a, double b)
    {
        return a + b;
    }
    public double sub(double a, double b)
    {
        return a - b;
    }
    public double mul(double a, double b)
    {
        return a * b;
    }
    public double div(double a, double b)
    {
        return a / b;
    }
    public CompositeType GetDataUsingDataContract(CompositeType composite)
    {
        if (composite == null)
        {
            throw new ArgumentNullException("composite");
        }
        if (composite.BoolValue)
        {
            composite.StringValue += "Suffix";
        }
        return composite;
    }
}
```



Add Service Reference?×

To see a list of available services on a specific server, enter a service URL and click Go. To browse for available services, click Discover.

Address:

Go | Discover▼

Services:

Operations:

▶  Service.svc

Select a service contract to view its operations.

1 service(s) found in the solution.

Namespace:

Advanced...OKCancel

WebForm.cs:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

public partial class _Default : System.Web.UI.Page
{
    ServiceReference1.ServiceClient service = new ServiceReference1.ServiceClient();

    protected void Page_Load(object sender, EventArgs e)
    {
    }

    protected void Button1_Click(object sender, EventArgs e)
    {
        double a = Convert.ToDouble(TextBox1.Text);
        double b = Convert.ToDouble(TextBox2.Text);
        double result = service.add(a, b);
        Label1.Text = "Addition" + result.ToString();
    }

    protected void Button2_Click(object sender, EventArgs e)
    {
        double a = Convert.ToDouble(TextBox1.Text);
        double b = Convert.ToDouble(TextBox2.Text);
        double result = service.sub(a, b);
        Label1.Text = "Subtraction" + result.ToString();
    }

    protected void Button3_Click(object sender, EventArgs e)
    {
        double a = Convert.ToDouble(TextBox1.Text);
        double b = Convert.ToDouble(TextBox2.Text);
        double result = service.mul(a, b);
        Label1.Text = "Multiplication" + result.ToString();
    }

    protected void Button4_Click(object sender, EventArgs e)
    {
        double a = Convert.ToDouble(TextBox1.Text);
        double b = Convert.ToDouble(TextBox2.Text);
        double result = service.div(a, b);
        Label1.Text = "Division" + result.ToString();
    }
}
```

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default.aspx.cs"
Inherits="_Default" %>
```

```
<!DOCTYPE html>
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
```

```
<head runat="server">
```

```
<title></title>
```

```
</head>
```

```
<body>
```

```
<form id="form1" runat="server">
```

```
<div>
```

```
Calculator using WCF<br />
```

```
<br />
```

```
First Number :<asp:TextBox ID="TextBox1" runat="server" style="margin-left:
44px" Width="171px"></asp:TextBox>
```

```
<br />
```

```
Second Number :
```

```
<asp:TextBox ID="TextBox2" runat="server" style="margin-left: 19px"
Width="171px"></asp:TextBox>
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<asp:Button ID="Button1" runat="server" Height="39px" OnClick="Button1_Click"
Text="ADD" Width="88px" />
```

```
<asp:Button ID="Button2" runat="server" Height="39px" OnClick="Button2_Click"
Text="SUB" Width="88px" />
```

```
<asp:Button ID="Button3" runat="server" Height="39px" OnClick="Button3_Click"
Text="MUL" Width="88px" />
```

```
<asp:Button ID="Button4" runat="server" Height="39px" OnClick="Button4_Click"
Text="DIV" Width="88px" />
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<asp:Label ID="Label1" runat="server" Text="RESULT :"></asp:Label>
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<br />
```

```
<br />
```

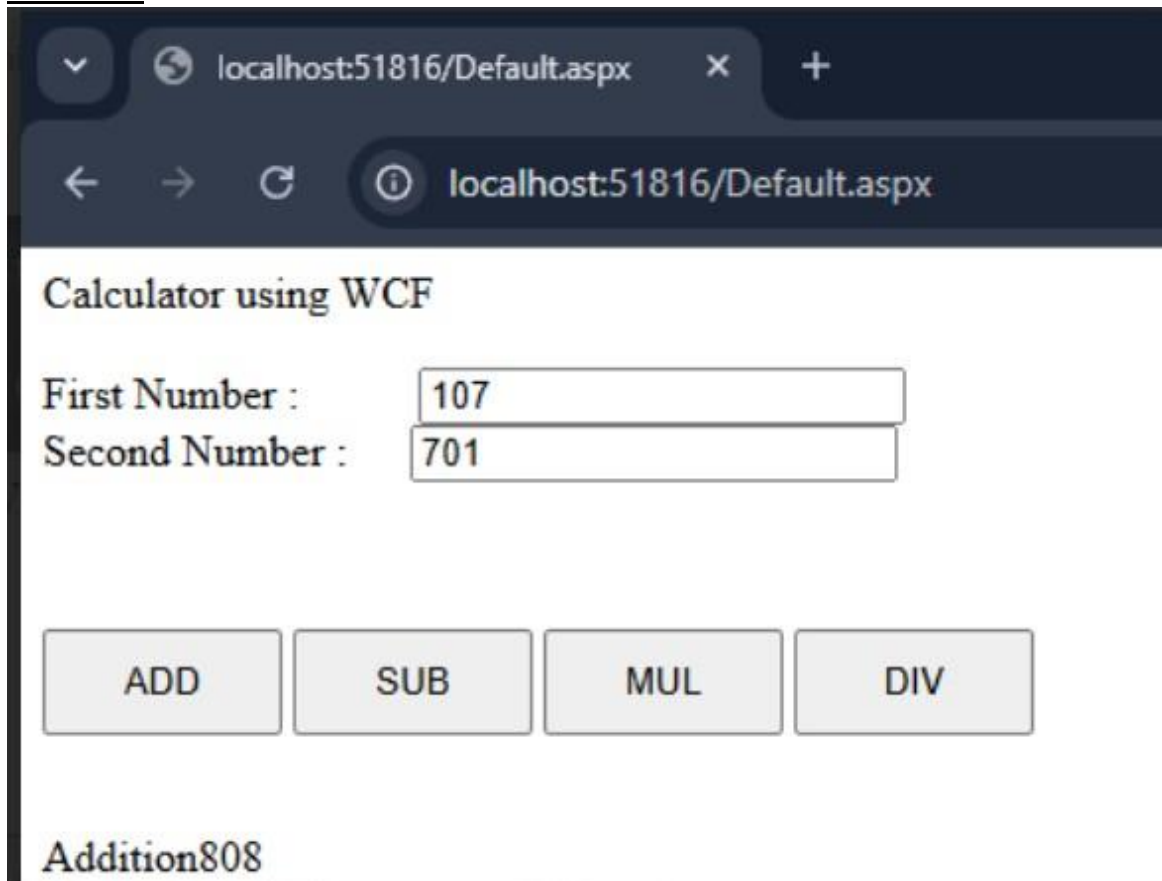
```
<br />
```

```
</div>
```

```
</form>
```

```
</body>
```

```
</html>
```

OUTPUT:

localhost:51816/Default.aspx

localhost:51816/Default.aspx

Calculator using WCF

First Number :

Second Number :

ADD SUB MUL DIV

Addition808

PRACTICAL NO. 16: Design web application using MVC framework.**Step 1: Create a Database in SQL Server**

Open SQL Server Management Studio (SSMS) or Visual Studio SQL Server Object Explorer.

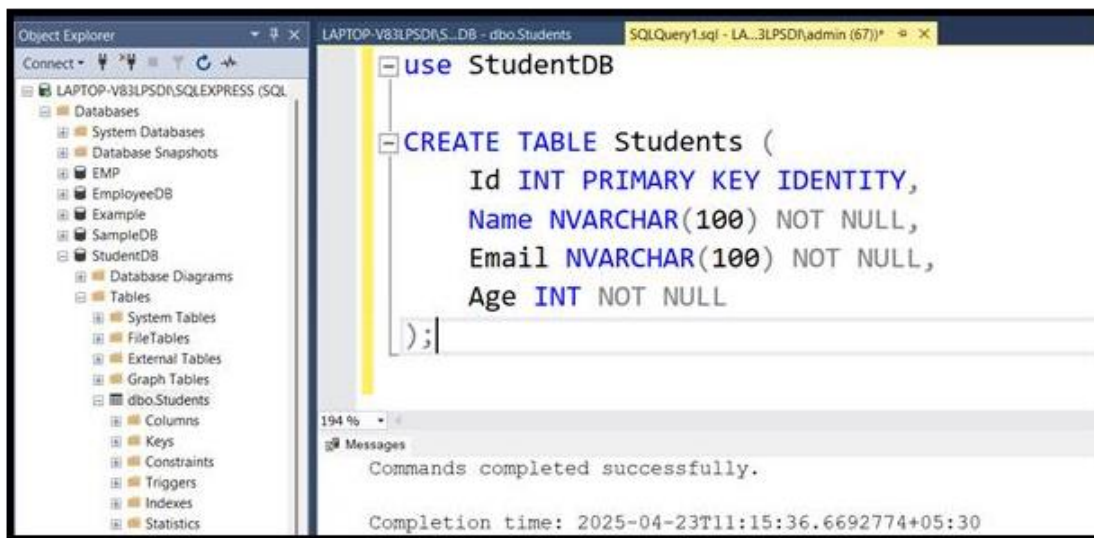
Create a Database named StudentDB.

Create a Table using the following SQL: Create Database StudentDB

use StudentDB

```
CREATE TABLE Students (  
    Id INT PRIMARY KEY IDENTITY,  
    Name NVARCHAR(100) NOT NULL,  
    Email NVARCHAR(100) NOT NULL,  
    Age INT NOT NULL
```

```
);
```



The screenshot shows the SQL Server Management Studio interface with the 'Students' table selected. The table contains three rows of data: a row with Id 1, Name 'ganesh', Email 'abc@gmail...', and Age 38; a row with Id 2, Name 'nilesh', Email 'xyz@gmail...', and Age 36; and a row with Id NULL, Name NULL, Email NULL, and Age NULL.

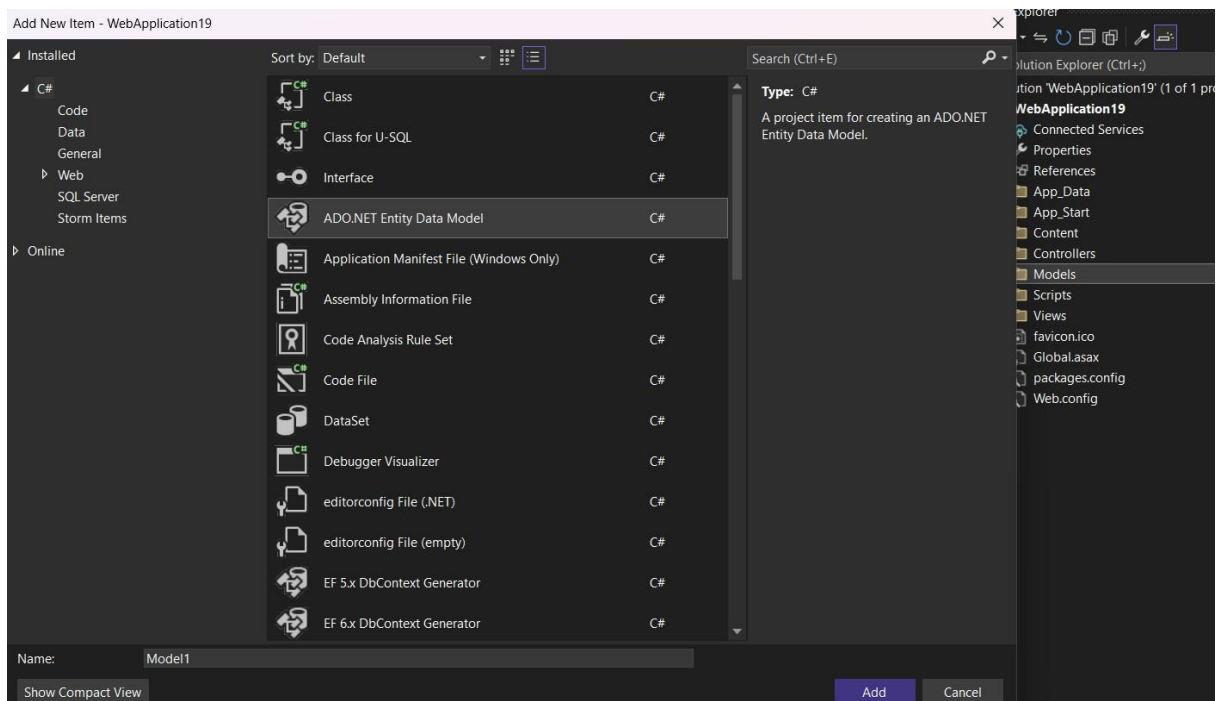
Id	Name	Email	Age
1	ganesh	abc@gmail...	38
2	nilesh	xyz@gmail...	36
NULL	NULL	NULL	NULL

Step 2: Create a New ASP.NET MVC Project

1. Open **Visual Studio**
2. Select **Create a new project**
3. Choose: **ASP.NET Web Application (.NET Framework)**
4. Name: StudentMVCApp
5. Choose **MVC** as the template
6. Click **Create**

Step 3: Add Entity Framework Model

1. Right-click the **Models** folder → Add → New Item
2. Choose **ADO.NET Entity Data Model**
3. Name it: StudentModel.edmx
4. Choose: "EF Designer from database"
5. Select your SQL Server database (StudentDB)
6. Select the Students table
7. Finish to generate model classes



Entity Data Model Wizard

 **Choose Model Contents**

What should the model contain?

 **EF Designer from database**

 Empty EF Designer model

 Empty Code First model

 Code First from database

Creates a model in the EF Designer based on an existing database. You can choose the database connection, settings for the model, and database objects to include in the model. The classes your application will interact with are generated from the model.

< Previous **Next >** Finish Cancel

Connection Properties

Enter information to connect to the selected data source or click "Change" to choose a different data source and/or provider.

Data source:
Microsoft SQL Server (SqlClient) Change...

Server name:
LAPTOP-V83LPSPDI\SQLEXPRESS Refresh

Log on to the server

Authentication: Windows Authentication

User name:

Password:

Encrypt: Mandatory (True)

☒ Trust Server Certificate

☐ Save my password

Connect to a database

☒ Select or enter a database name:
StudentDB

☐ Attach a database file:
 Browse...

Logical name:

Advanced...

Test Connection OK Cancel

Entity Data Model Wizard



Choose Your Data Connection

Which data connection should your application use to connect to the database?

laptop-v83lpsdi\sqlexpress.StudentDB.dbo

New Connection...

This connection string appears to contain sensitive data (for example, a password) that is required to connect to the database. Storing sensitive data in the connection string can be a security risk. Do you want to include this sensitive data in the connection string?

☐ No, exclude sensitive data from the connection string. I will set it in my application code.

☐ Yes, include the sensitive data in the connection string.

Connection string:

metadata=res://*/Models.Model1.csdl|res://*/Models.Model1.ssdl|res://*/Models.Model1.msl;provider=System.Data.SqlClient;provider connection string="data source=LAPTOP-V83LPSDI\SQLEXPRESS;initial catalog=StudentDB;integrated security=True;trustservercertificate=True;MultipleActiveResultSets=True;App=EntityFramework"

☒ Save connection settings in Web.Config as:

StudentDBEntities

< Previous

Next >

Finish

Cancel

Entity Data Model Wizard



Choose Your Version

Which version of Entity Framework do you want to use?

☒ Entity Framework 6.x

☐ Entity Framework 5.0

 It is also possible to install and use other versions of Entity Framework.

[Learn more about this](#)

< Previous

Next >

Finish

Cancel

Entity Data Model Wizard



Choose Your Database Objects and Settings

Which database objects do you want to include in your model?

☒ Tables

☒ > dbo

☐ Views

☐ Stored Procedures and Functions

☒ Pluralize or singularize generated object names

☒ Include foreign key columns in the model

☐ Import selected stored procedures and functions into the entity model

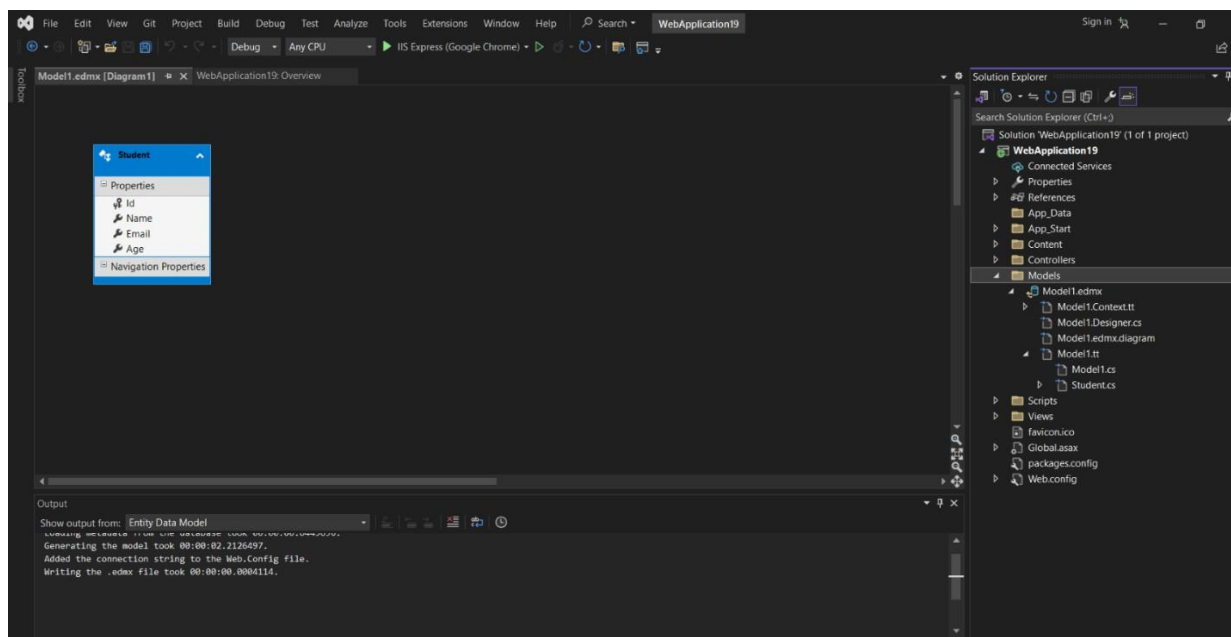
Model Namespace:

< Previous

Next >

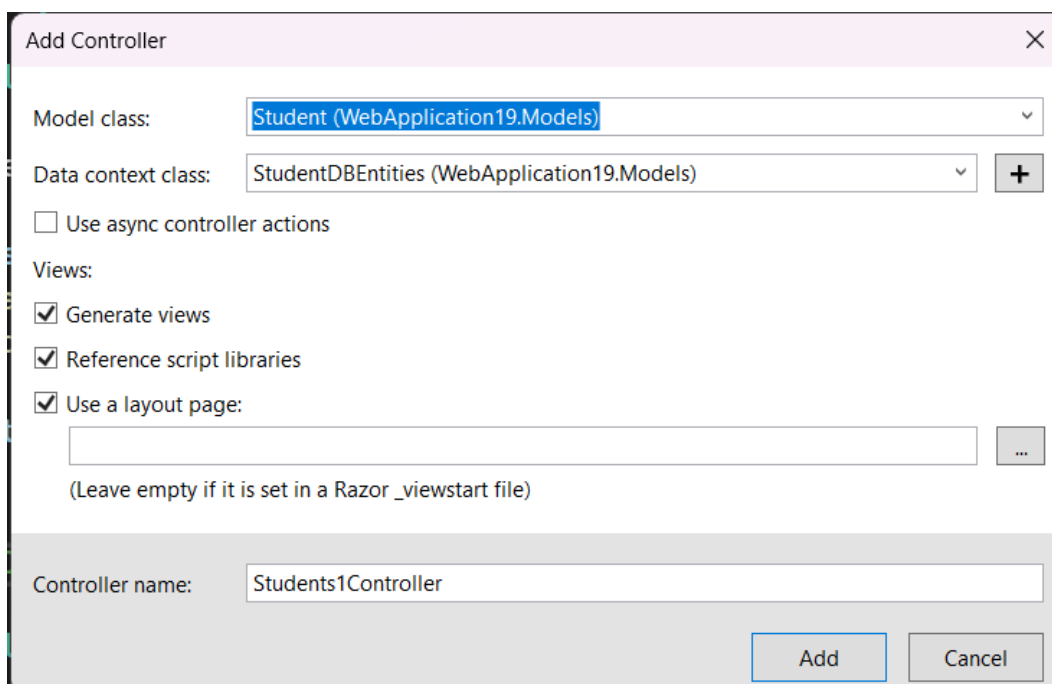
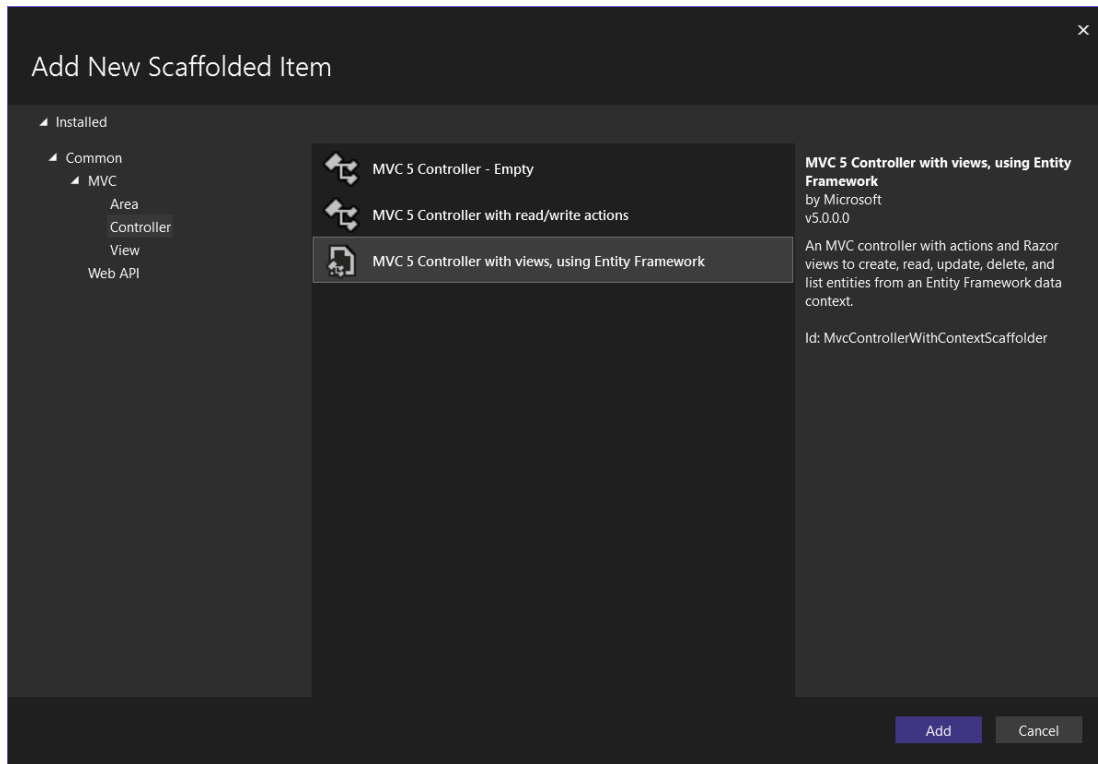
Finish

Cancel



Step 4: Create Controller

1. Right-click **Controllers** → Add → Controller
2. Choose: **MVC 5 Controller with views, using Entity Framework**
3. Model class: Student
4. Data context: StudentDBEntities (if using .edmx) or StudentDbContext
5. Click **Add**



Step 5: Create Views

You can right-click on each controller action and choose **Add View**, or use the auto-generated ones.

Example: Views/Students/Index.cshtml

```
@model IEnumerable<StudentMVCApp.Models.Student>
@{
    ViewBag.Title = "Student List";
}
<h2>Student List</h2>
<p>@Html.ActionLink("Create New", "Create")</p>
<table class="table">
    <tr>
        <th>@Html.DisplayNameFor(model => model.Name)</th>
        <th>@Html.DisplayNameFor(model => model.Email)</th>
        <th>@Html.DisplayNameFor(model => model.Age)</th>
        <th>Actions</th>
    </tr>
    @foreach (var item in Model) {
        <tr>
            <td>@item.Name</td>
            <td>@item.Email</td>
            <td>@item.Age</td>
            <td>
                @Html.ActionLink("Edit", "Edit", new { id = item.Id }) |
                @Html.ActionLink("Details", "Details", new { id = item.Id }) |
                @Html.ActionLink("Delete", "Delete", new { id = item.Id })
            </td>
        </tr>
    }
</table>
```

Step 6: Set Default Route

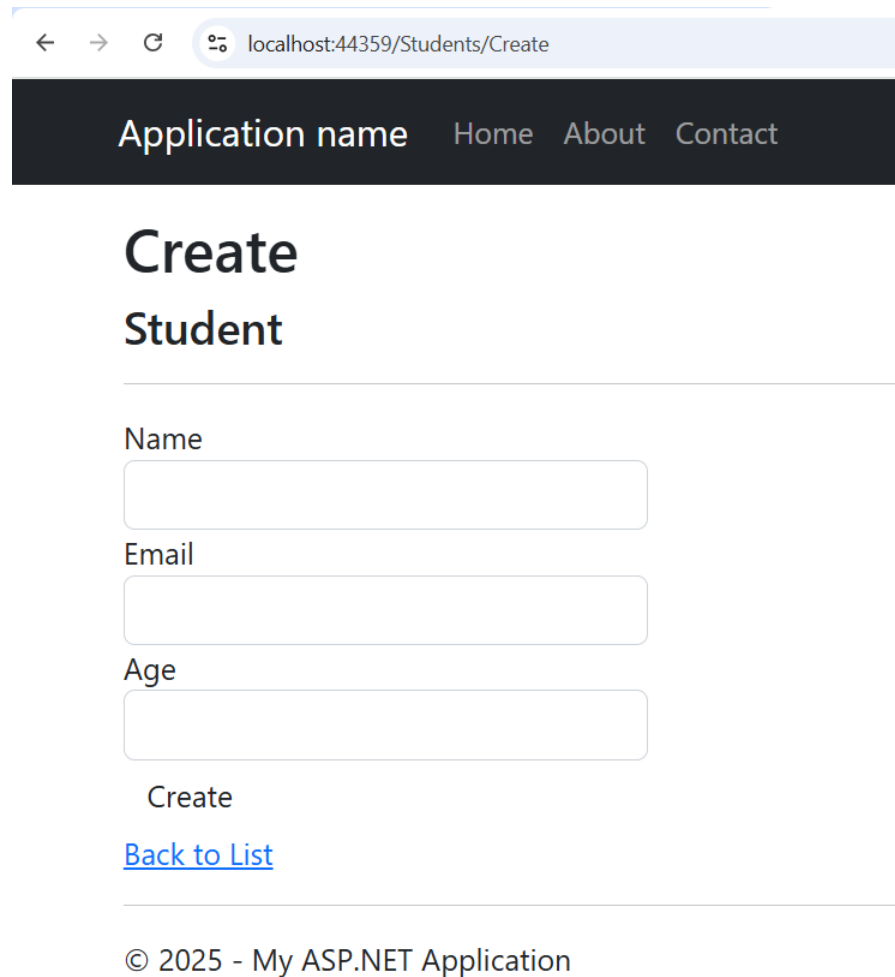
In App_Start/RouteConfig.cs, change default route to:

csharp

CopyEdit

```
defaults: new { controller = "Students", action = "Index", id = UrlParameter.Optional }
```

OUTPUT:



← → ↻ 📄 localhost:44359/Students/Create

Application name Home About Contact

Create Student

Name

Email

Age

Create

[Back to List](#)

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